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Controllability of some Deterministic and Stochastic Systems of Parabolic Equations

Par

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0.1 Introduction

Ce mémoire de l'habilitation à diriger des recherches est l'occasion d'exposer mon parcours d'enseignant-chercheur à l'école normale supérieure université Cadi Ayyad et de la mise en valeur des missions qui m'ont été confiées au sein du ministère de l'éducation nationale avant mon concours de professeur assistant.

Mon axe de recherche est toujours l'étude de la contrôlabilité des systèmes d'équations couplés. En effet, nous considérons le système parabolique a n équations dégénérées soumis a m contrôles

$$\begin{cases} \partial_t y - D\mathcal{M}y + Ay = Bv\mathbb{1}_\omega & \text{in } Q_T, \\ Cy = 0, & \text{on } \Sigma_T, \\ y(0, \cdot) = y_0, & \text{in } (0, 1), \end{cases} \quad (\text{I})$$

où D est la matrice de diffusion, A matrice de couplage, B est la matrice de contrôle et $\mathcal{M}$ est l'opérateur différentiel défini par $\mathcal{M}y = (a(x)y_x)_x$.

Au cours de ma thèse de doctorat, nous avons traité la contrôlabilité du système (I) dans les cas suivants:

- le cas d'un système cascade soumis à plusieurs contrôles et une matrice de diffusion de la forme $D = \text{diag}(d_1, \dots, d_n)$ avec $(d_i > 0)$ voir [1].
- le cas d'un système complet où la matrice de diffusion $D = d.I_n$, nous avons montré que la condition algébrique de Kalman $\text{rank}[A|B] = n$ caractérise la contrôlabilité à zéro du système (I) Voir [2].

Dans le présent travail nous nous intéressons au cas ou la matrice de diffusion D n'est pas diagonalisable, nous montrons également si la condition algébrique de Kalman $\text{rank}[A|B] = n$ est satisfait par les deux matrices de couplage A et de contrôle B le système (I) est contrôlable à zéro (voir la sous-section 0.3.1 et [3]).

Passant maintenant dans le contexte stochastique, ou nous considérons le systèmes des équations stochastiques rétrogrades couplées suivantes

$$\begin{cases} d_t y_1 = \left[-\Delta y_1 + \sum_{i=1}^m a_{1i} y_i + c_1 Y_1 + \mathbb{1}_{G^v} \right] dt + Y_1 dW(t), \\ d_t y_2 = \left[-\Delta y_2 + \sum_{i=1}^m a_{2i} y_i + c_2 Y_2 \right] dt + Y_2 dW(t), \\ \vdots \\ d_t y_m = \left[-\Delta y_m + \sum_{i=1}^m a_{mi} y_i + c_m Y_m \right] dt + Y_m dW(t), \\ y_i = 0 \quad 1 \leq i \leq m & \text{on } \Sigma, \\ y_i(T, \cdot) = y_{iT}(\cdot) \quad 1 \leq i \leq m & \text{in } \mathcal{D}, \end{cases} \quad \text{in } Q \quad (\text{II})$$

où (y_i, Y_i) est la variable d'état de la i -ième équations.

Nous montrons si la première matrice de couplage est triangulaire supérieure avec

une sous diagonale non nul, alors le système (II) peut être contrôlé avec un seul en utilisant un seul contrôle (voir la sous-section 0.3.2 et [4]).

Et finalement l'objet du troisième papier est l'étude du problème stochastique rétrograde suivant

$$\begin{cases} d_t y_1 = [-\Delta y_1 + a_1 y_1 + c_{1,1} Y_1 + c_{1,2} Y_2 + \mathbb{1}_G v] dt + Y_1 dW(t), & \text{in } Q \\ d_t y_2 = [-\Delta y_2 + a_2 y_2 + c_{2,1} Y_1 + c_{2,2} Y_2] dt + Y_2 dW(t), & \text{in } Q \\ y_i = 0 \quad i \in \{1, 2\} & \text{on } \Sigma, \\ y_i(T, \cdot) = y_{iT}(\cdot) \quad i \in \{1, 2\} & \text{in } \mathcal{D}, \end{cases} \quad (\text{III})$$

où les deux équations sont couplés uniquement à l'aide de leurs seconde variables d'état. La condition suivante

$$c_{21} \geq c > 0 \quad \text{or} \quad -c_{21} \geq c > 0 \quad \text{in } G_0 \times (0, T). \quad (0.1.1)$$

sur la seconde matrice de couplage nous permet de prouver la contrôlabilité à zéro du système (III) sous l'action d'un seul contrôle (voir la sous-section 0.3.3 et [5]). Ce mémoire est présenté comme suit : section 2 est dédié à mon curriculum vitae. Section 3 est consacré à mes récents travaux de recherches. Dans la section 4 je présente mon parcours professionnel est pédagogique et enfin je conclus ce travail par quelques perspectives.

0.2 Curriculum Vitae

0.2.1 Information personnelles

- **Mohamed FADILI**
- **Date et lieu de naissance** : 08 Novembre 1970 à had bou moussa fquib ben saleh.
- **Situation familiale** : Marié.
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- **Adresse Personnelle** : Ibn khaldoun 1 Imm C numéro 4 Massira 3 Marrakech.
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0.2.2 Cursus Universitaire

- Doctorat en Mathématiques, Université Cadi Ayyad, Marrakech, 2018
- Master Mathématiques et applications, Université Cadi Ayyad, Marrakech, 2013
- Agrégation Mathématiques 2002
- C.E.A Calcul Formel, Université Cadi Ayyad, Marrakech, 1994

- Licence Es-Sciences en Mathématiques, Université Cadi Ayyad, 1993
- Baccalauréat Sciences, Lycée Ibn Sina, Beni-Mellal, 1989.

0.2.3 Liste des Publications

- [1] M. Fadili and L. Maniar, *Null controllability of n -coupled degenerate parabolic systems with m -controls*, J. Evol. Equ. (2017), no. 4, 1-30.
- [2] E. M. Ait Ben Hassi, M. Fadili and L. Maniar, *On Algebraic condition for null controllability of some coupled degenerate systems*. *Mathematical Control and Related Fields* (2018) doi:10.3934/mcrf.2019004.
- [3] E. M. Ait Ben Hassi, M. Fadili and L. Maniar, *Controllability of a system of degenerate parabolic equations with non-diagonalizable diffusion matrix*. *Mathematical Control and Related Fields* (2020)(10) no.3, 623–642. doi:10.3934/mcrf.2020013.
- [4] Mohamed Fadili, *Controllability of a backward stochastic system of coupled parabolic heat equations by one control force*. *Evolution Equation and Control Theory* (2022) doi:103934/eect.2022037.
- [5] M. Fadili and L. Maniar, *Controllability of two stochastic backward parabolic heat equations coupled by their second state variable under one control force*. submitted to Acta Applicandae Mathematicae.
- [6] M. Baroun, M. Fadili, A. Khchine and L. Maniar, *Carleman estimates and controllability of stochastic degenerate parabolic heat equations*. preprint, arXiv:2202.10070, submitted to EJC (2022).
- [7] E. M. Ait Ben Hassi, M. Fadili and L. Maniar, *Null controllability of a degenerate parabolic equation with one finite delay*. submitted to The Electronic Journal of Qualitative Theory of Differential Equations(EJQTDE).

0.2.4 Liste des Communications et Conférences

12 Septembre 2022 : *Controllability of Two Stochastic Backward Parabolic Heat Equations Coupled by their Second State Variable under One Control Force* . the 12th Euro-Maghrebian Workshop on Evolution Equations and COST WG1-Meeting held on September 12-16, 2022 at Agadir.

26 Mars 2022 : *Controllability of a stochastic parabolic degenerate equation under the action of two controls force*. International Conference on Research in Applied Mathematics and Computer Science ICRAMS'2022

20 Juillet 2020 : M. Baroun, M. Fadili, A. Khchine and L. Maniar *Null controllability for degenerate backward stochastic equations*. journée de la Théorie d'Analyse et Control des systèmes infinis TACSI'20, Marrakech, Morocco.

30 Avril 2019 : Participation à la journée thématique "Enseignement des mathématiques -Approches et pratiques innovantes-" organisé par l'Ecole Normale Supérieure de Marrakech (2019).

3-5 Avril 2019 : Participation à la réunion scientifique "UMMISCO " au centre UMMISCO Méditerranée, Université Cadi Ayyad. Marrakech (2019).

29-30 Mars 2019 : *Null controllability of a degenerate parabolic equation with delay.* The first International Conference in Applied Mathematics and Computer Science ICRAMCS 2019. Faculté des Sciences Ben M'sick Casablanca (2019).

27 Mars 2019 : Participation à la journée thématique "AUTOUR DES MATHÉMATIQUES " organisé par l'Ecole Hassania des Travaux Publics. Casablanca (2019).

4-6 Avril 2018 : *On null controllability of some degenerate parabolic systems.* The international conference "Humboldt Kolleg ". Marrakech (2018).

1 Mars 2018 : Participation à la journée thématique sur les problèmes paraboliques dégénérés. Faculté des sciences et techniques Université Hassan Premier. Settat (2018).

19-23 Juin 2017 : Participation à la conférence: 20th Internet Seminar qui a eu lieu à Baronissi (Italie).

22 Avril 2017 : *La contrôlabilité à zéro des systèmes dégénéré paraboliques couplés,* lors de la cinquième Journée des Sciences de l'Ingénieur JSI Casablanca (Maroc)

16-18 Mars 2017 : *Null controllability of some coupled weak degenerate systems,* au 5ème Congrès International de la SM2A Meknès (Maroc)

23 Avril 2016 : *Null controllability for a cascade degenerate system with one control force,* La quatrième Journée des Sciences de l'Ingénieur, Faculté des Sciences Ben M'sik, Casablanca.

12-16 Decembre 2016 : Participation à la conférence: Marrakesh workshop on Control, Inverse Problems and Stabilization of infinite Dimensional Systems.

10-12 Novembre 2015 : *Controllability results for cascade systems of m coupled degenerate parabolic PDEs by one control force,* Conférence Internationale de Mathématiques Appliquées de Safi

21 Juin 2014 : Participation à la troisième Journée des Sciences de l'Ingénieur, Faculté des Sciences Ben M'sik, Casablanca.

22-26 Septembre 2014 : Participation à la conférence: 9th Euro-Magrebien Work-

shop of Evolution Equations, Marrakesh

0.2.5 Certificats

26 Mars 2022 : “chairman of the session of dynamical systems and control ”ICRAMS’2022

24-26 Mars 2022 : “Membre du Comité Scientifique du la quatrième édition (The International Conference on Research in Applied Mathematics and Computer Science) ”. ICRAMS’2022

Mars-Juin 2021 :Pédagogie Universitaire **UCA**

08 juillet 2017 : TIC National

27 Mars 2015 :Office PowerPoint 2010

16 Decembre 2014 :Office Excel 2010, Office Word 2010 .

0.3 Activités de Recherche

0.3.1 Controllability of a system of degenerate parabolic equations with non-diagonalizable diffusion matrix

CONTROLLABILITY OF A SYSTEM OF DEGENERATE PARABOLIC EQUATIONS WITH NON-DIAGONALIZABLE DIFFUSION MATRIX

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(Communicated by Piermarco Cannarsa)

ABSTRACT. In this paper we study the null controllability of some non diagonalizable degenerate parabolic systems of PDEs, we assume that the diffusion, coupling and controls matrices are constant and we characterize the null controllability by an algebraic condition so called *Kalman's rank* condition.

1. Introduction and main result. In this paper we focus on the controllability properties of the following non-diagonalizable parabolic degenerate system

$$\begin{cases} \partial_t Y = (\mathbf{D}\mathcal{M} + A)Y + Bv\mathbb{1}_\omega & \text{in } Q, \\ \mathbf{C}Y = 0 & \text{on } \Sigma, \\ Y(0, x) = Y_0(x) & \text{in } (0, 1), \end{cases} \quad (1)$$

where $Q := (0, T) \times (0, 1)$, $\Sigma := (0, T) \times \{0, 1\}$, for $T > 0$ and $\omega \subset (0, 1)$ is a (small) nonempty open control region, $\mathbb{1}_\omega$ denotes the characteristic function of ω . The diffusion matrix $\mathbf{D}$ is a non-diagonalizable $n \times n$ matrix that satisfies the following assumptions :

- there exists $\alpha_0 > 0$ such that

$$\mathbf{D}\xi \cdot \xi \geq \alpha_0 |\xi|^2 \quad \forall \xi \in \mathbb{R}^n \quad (2)$$

- there exists a non-singular matrix

$$P \in \mathcal{L}(\mathbb{C}^n) \text{ such that } \mathbf{D} = PJP^{-1} \quad (3)$$

for some $J \in \mathcal{L}(\mathbb{C}^n)$ of the form

$$J = \text{diag}(J_1, \dots, J_p),$$

2010 *Mathematics Subject Classification.* Primary: 35K20, 35K65, 47D06; Secondary: 93B05, 93B07.

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where the J_i are the Jordan blocks associated to the eigenvalues d_i of $\mathbf{D}$.

$$J_i = \begin{bmatrix} d_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & d_i \end{bmatrix} \quad (4)$$

with $\operatorname{Re} d_i > 0$.

The coupling matrix A is a $n \times n$ constant matrix and the control matrix B is a $n \times m$ constant matrix. The operator $\mathcal{M}$ is defined by $\mathcal{M}y = (ay_x)_x$ for $y \in D(\mathcal{M}) \subset L^2(0, 1)$. For $Y = (y_1, \dots, y_n)^*$, $\mathcal{M}Y$ denotes $(\mathcal{M}y_1, \dots, \mathcal{M}y_n)^*$. The function a is a diffusion coefficient which degenerates at 0 (i.e., $a(0) = 0$) and which can be either weak degenerate (WD), i.e.,

$$(WD) \begin{cases} (i) \ a \in \mathcal{C}([0, 1]) \cup \mathcal{C}^1((0, 1]), \ a > 0 \text{ in } (0, 1], \ a(0) = 0, \\ (ii) \ \exists K \in [0, 1) \text{ such that } xa'(x) \leq Ka(x), \ \forall x \in [0, 1], \end{cases} \quad (5)$$

or strong degenerate (SD), i.e.,

$$(SD) \begin{cases} (i) \ a \in \mathcal{C}^1([0, 1]), \ a > 0 \text{ in } (0, 1], \ a(0) = 0, \\ (ii) \ \exists K \in [1, 2) \text{ such that } xa'(x) \leq Ka(x) \ \forall x \in [0, 1], \\ (iii) \ \begin{cases} \exists \theta \in (1, K]x \mapsto \frac{a(x)}{x^\theta} \text{ is nondecreasing near } 0, \text{ if } K > 1, \\ \exists \theta \in (0, 1)x \mapsto \frac{a(x)}{x^\theta} \text{ is nondecreasing near } 0, \text{ if } K = 1. \end{cases} \end{cases} \quad (6)$$

The boundary condition $\mathbf{C}Y = 0$ is either $Y(0) = Y(1) = 0$ in the weak degenerate case (WD) or $Y(1) = (aY_x)(0) = 0$ in the strongly degenerate case (SD).

It will be said that the system (1) is *null-controllable* at time T if, for any $Y_0 \in (L^2(0, 1))^n$, there exists $v \in L^2((0, T) \times \omega)^m$ such that the associated solution satisfies

$$y(T, x) = 0 \text{ in } (0, 1).$$

The system (1) is said to be *approximately controllable* at time T if, for any $Y_0, Y_1 \in L^2((0, 1))^n$ and $\varepsilon > 0$, there exists $v \in L^2((0, T) \times \omega)^m$ such that the solution of (1) corresponding to the initial condition Y_0 satisfies

$$\|y(T, x) - Y_1\|_{(L^2(0, 1))^n} \leq \varepsilon.$$

Controlling coupling systems of partial differential equations attracted growing interest during the last decade, the main question is whether it is possible to control such systems with fewer controls (i.e the number of controls is less than the number of equations). For finite dimensional linear systems, the controllability can be characterized by algebraic rank condition on the matrices generating the dynamics and taking account of the control action. The theory has been adapted and extended to more general systems including infinite dimensional systems. At our knowledge, in the nondegenerate case, M. Gonzalez-Burgos, L. de Teresa [18] provided a null controllability result for a cascade parabolic system by one control force under a condition on the sub-diagonal of the coupling matrix. F. Ammar Khodja et al. [5, 6] obtained several results characterizing the null controllability of fully coupled systems with m -control forces by a generalized Kalman rank condition. In [16], the authors gave controllability results for a system in the case where the diffusion matrix is non diagonalizable.

For degenerate systems, the case of two coupled equations ($n = 2$), cascade systems are considered in [13, 14] and in [1, 2] the authors have studied the null controllability of degenerate noncascade parabolic systems.

In [15], we have extended the null controllability results obtained by Ammar Khodja et al. [6] to a class of parabolic degenerate systems of PDEs in the two following cases

1. the coupling matrix A is a cascade one and the diffusion matrix $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ where $d_i > 0$, $i = 1, \dots, n$,
2. the coupling matrix A is a full matrix (noncascade) and the diffusion matrix $\mathbf{D} = dI_n$ with $d > 0$.

On the other hand, in [3] we studied the null controllability of system (1) in the case where the diffusion matrix $\mathbf{D}$ is diagonalizable $n \times n$ matrix with positive real eigenvalues, i.e.,

$$\mathbf{D} = P^{-1} \mathbf{J} P, \quad P \in \mathcal{L}(\mathbb{R}^n), \quad \det(P) \neq 0, \quad (7)$$

where $\mathbf{J} = \text{diag}(d_1, \dots, d_n)$, $d_i > 0$, $1 \leq i \leq n$.

In the current paper, we assume that diffusion matrix $\mathbf{D}$ is non-diagonalizable. We use the same approach as [16] without imposing that Jordan's block sizes are bounded by 4. Thus our proof is also an improvement of the one given in [16] for the nondegenerate case.

Notice that, if the condition (2) is satisfied, for every $v \in L^2((0, T) \times \omega; \mathbb{R}^m)$ and every $y_0 \in L^2((0, 1); \mathbb{R}^n)$, the system (1) possesses a unique weak solution y , with $y \in C([0, T], (L^2(0, 1))^n) \cap L^2(0, T; (H_a^1)^n)$ see Section 2.

In order to study the null controllability of system (1), we will consider the following corresponding adjoint problem

$$\begin{cases} -z_t - \mathbf{D}^* \mathcal{M} z = A^* z & \text{in } Q, \\ \mathbf{C} z = 0 & \text{on } \Sigma, \\ z(T, x) = z_T(x) & \text{in } (0, 1). \end{cases} \quad (8)$$

Since the null controllability of system (1) is equivalent to the existence of a positive constant C such that, for every $z_T \in (L^2(0, 1))^n$, the solution $z \in C^0([0, T]; (L^2(0, 1))^n)$ to the adjoint system (8) satisfies the observability inequality:

$$\|z(\cdot, 0)\|_{(L^2(0, 1))^n}^2 \leq C \iint_{(0, T) \times \omega} |B^* z(x, t)|^2, \quad (9)$$

The strategy used in this case is slightly different from the one used in [3], although in both cases it is necessary to show Carleman estimates for a scalar PDE of order $2n$ in space.

Note that this technique failed in the case where $\mathcal{M}$ is a compact operator, since its spectrum admits zero as a point of accumulation, so the Kalman condition is no longer verified, which does not ensure a perfect coupling of the equations.

All along the article, we use generic constants for the estimates, whose values may change from line to line.

Let us remark that when $A \in \mathcal{L}(\mathbb{R}^n)$ and $B \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ are constant matrices, $[A|B] \in \mathcal{L}(\mathbb{R}^{nm}, \mathbb{R}^n)$ is the matrix given by

$$[A|B] = (A^{n-1}B | A^{n-2}B | \dots | AB | B)$$

With this notation, we have the following main result.

Theorem 1.1. *Let assume that $\mathbf{D}$, $A \in \mathcal{L}(\mathbb{R}^n)$, $B \in \mathcal{L}(\mathbb{R}^m; \mathbb{R}^n)$ such that $\mathbf{D}$ satisfies the assumptions (2)-(4). Then system (1) is null controllable at time T if and only if*

$$\text{rank}[\lambda_i \mathbf{D} - A|B] = n \quad \forall i \geq 1. \quad (10)$$

The rest of this paper is organized as follows. In Section 2, we prove the well-posedness of the problem (1). Section 3 is devoted to some controllability results for one parabolic equation. In Section 4, we prove some useful estimates under the assumption (10) and we establish Carleman estimates for a scalar PDE of order $2n$ in space. In Section 5, we give the proof of the main result. And finally, in Section 6 we study the null controllability of semilinear systems.

2. Wellposedness of the problem. The semigroup generated by the operator $(\mathcal{M}, D(\mathcal{M}))$ is analytic with angle $\frac{\pi}{2}$ (see [9, Theorem 2.8]). In order to prove the wellposedness of the problem (1), it suffices to show that the operator $\mathbf{D}\mathcal{M}$ generates a c_0 -semigroup. In fact, similarly like in [22], under the assumption that all eigenvalues of the diffusion matrix $\mathbf{D}$ have positive real part, we prove that $\mathbf{D}\mathcal{M}$ is the generator of an analytic semigroup. Let us first introduce the following weighted spaces :

$$H_a^1 = \{u \in L^2(0, 1)/u \text{ absolutely continuous in } [0, 1], \\ \sqrt{a}u_x \in L^2(0, 1) \text{ and } u(1) = u(0) = 0\}$$

and

$$H_a^2 = \{u \in H_a^1(0, 1)/au_x \in H^1(0, 1)\}.$$

in the (WD) case and

$$H_a^1 = \{u \in L^2(0, 1)/u \text{ absolutely continuous in } (0, 1], \sqrt{a}u_x \in L^2(0, 1) \text{ and } u(1) = 0\}$$

and

$$H_a^2 = \{u \in H_a^1(0, 1)/au_x \in H^1(0, 1)\} \\ = \{u \in L^2(0, 1)/u \text{ absolutely continuous in } (0, 1], au \in H_0^1(0, 1), \\ au_x \in H^1(0, 1) \text{ and } (au_x)(0) = 0\},$$

in the (SD) case with the norms

$$\|u\|_{H_a^1}^2 = \|u\|_{L^2(0, 1)}^2 + \|\sqrt{a}u_x\|_{L^2(0, 1)}^2, \quad \|u\|_{H_a^2}^2 = \|u\|_{H_a^1}^2 + \|(au_x)_x\|_{L^2(0, 1)}^2.$$

We recall also some results about the spectrum of the operator $-\mathcal{M}$ already used in [3], indeed, the operator $-\mathcal{M}$ is a definite positive operator. Knowing that $D(\mathcal{M}) = H_a^2(0, 1)$ is compactly embedded in $L^2(0, 1)$ see [10]. Thus, the spectrum of $-\mathcal{M}$ consists of eigenvalues

$$0 < \lambda_1 < \lambda_2 < \dots < \lambda_j < \dots \quad \text{with} \quad \lambda_j \longrightarrow +\infty \quad (11)$$

Therefore : There exists a complete orthonormal set $\{\mathbf{w}_j\}$ of eigenvectors of $-\mathcal{M}$.

For all $z \in D(-\mathcal{M})$ we have

$$-\mathcal{M}z = \sum_{j=1}^{+\infty} \lambda_j \langle z, \mathbf{w}_j \rangle \mathbf{w}_j = \sum_{j=1}^{+\infty} \lambda_j E_j z, \quad (12)$$

where $\langle \cdot, \cdot \rangle$ is the inner product in $L^2(0, 1)$ and

$$E_j z = \langle z, \mathbf{w}_j \rangle \mathbf{w}_j.$$

So, $\{E_j\}$ is a family of complete orthogonal projections in $L^2(0, 1)$ and

$$z = \sum_{j=1}^{+\infty} E_j z, \quad z \in L^2(0, 1).$$

$-\mathcal{M}$ generates a strongly continuous semigroup $\{e^{-\mathcal{M}t}\}$ given by

$$e^{-\mathcal{M}t} z = \sum_{j=1}^{+\infty} e^{\lambda_j t} E_j z.$$

In the Hilbert space $\mathbb{H} := (L^2(0, 1))^n$ we define the following linear operator: $\mathcal{A} : D(\mathcal{A}) \subset \mathbb{H} \rightarrow \mathbb{H}$ given by $D(\mathcal{A}) = (H_a^2)^n$ and $\mathcal{A}u = -\mathbf{D}\mathcal{M}u$.

We have the following result.

Theorem 2.1. *Assume that all eigenvalues of $\mathbf{D}$ have positive real part. Then $\mathcal{A}$ is sectorial and therefore, $-\mathcal{A}$ is the generator of an analytic semigroup $\{e^{-\mathcal{A}t} : t \geq 0\}$ in $\mathbb{H}$.*

Proof. Let $\theta \in (0, \frac{\pi}{2})$ such that $|\arg \lambda_{\mathbf{D}}| < \theta$ for any eigenvalue $\lambda_{\mathbf{D}}$ of $\mathbf{D}$. We prove that the sector

$$S = \{z \in \mathbb{C} : \theta \leq |\arg z| \leq \pi, z \neq 0\}$$

is in the resolvent set of $\mathcal{A}$ and there exists a constant C such that for any $z \in S$

$$\|(z - \mathcal{A})^{-1}\| \leq \frac{C}{|z|}$$

For $z \in S$ and $f \in \mathbb{H}$, let u be given by

$$u = \sum_{j=1}^{\infty} (z - \lambda_j \mathbf{D})^{-1} f_j \mathbf{w}_j$$

where $f_j = \int_{(0,1)} f \mathbf{w}_j dx$. Since $z \in S$ implies $\frac{z}{\lambda_j}$ is not an eigenvalue of $\mathbf{D}$, the matrix $z - \lambda_j \mathbf{D}$ is invertible and there exists a constant $C > 0$ such that $\|(z - \lambda_j \mathbf{D})^{-1}\| \leq \frac{C}{|z|}$ for all $j \geq 1$. It follows that the above series is convergent in $\mathbb{H}$, so u is well defined and $\|u\| \leq \frac{C}{|z|} \|f\|$. Also,

$$\begin{aligned} zu + \mathbf{D}\mathcal{M}u &= \sum_{j=1}^{\infty} [z(z - \lambda_j \mathbf{D})^{-1} f_j - \lambda_j \mathbf{D} f_j] \mathbf{w}_j \\ &= \sum_{j=1}^{\infty} (z - \lambda_j \mathbf{D})^{-1} [z - \lambda_j \mathbf{D}] f_j \mathbf{w}_j \\ &= \sum_{j=1}^{\infty} f_j \mathbf{w}_j = f, \end{aligned}$$

so, $u = (z - \mathcal{A})^{-1} f$. Therefore, z is in the resolvent set of $\mathcal{A}$,

$$\|(z - \mathcal{A})^{-1}\| \leq \frac{C}{|z|}$$

and the proof is complete. $\square$

Thus, system (1) is well posed in the sens of semigroup theory and the following global existence result holds.

Theorem 2.2. *Under the Hypothesis (3)-(4), for all $(y_1^0, \dots, y_n^0) \in (L^2(0, 1))^n$ and $v \in (L^2(Q))^m$ there exists a unique mild solution $(y_1(t), \dots, y_n(t))$ of system (1) which belongs to*

$$X_T := C\left([0, T], (L^2(0, 1))^n\right) \cap L^2\left(0, T; (H_a^1)^n\right)$$

and satisfies

$$\begin{aligned} \sup_{t \in [0, T]} \|(y_1, \dots, y_n)(t)\|_{(L^2(0, 1))^n}^2 + \int_0^T \|(\sqrt{a}y_{1x}, \dots, \sqrt{a}y_{nx})\|_{L^2}^2 dt \\ \leq C_T \left(\|(y_1^0, \dots, y_n^0)\|_{(L^2(0, 1))^n}^2 + \|v\|_{(L^2(Q))^m}^2 \right), \end{aligned}$$

for a constant $C_T > 0$.

Moreover, if $(y_1^0, \dots, y_n^0) \in (H_a^1)^n$, then

$$(y_1, \dots, y_n) \in C\left([0, T], (H_a^1)^n\right) \cap H^1\left(0, T; (L^2(0, 1))^n\right) \cap L^2\left(0, T; (H_a^2)^n\right)$$

and

$$\begin{aligned} \sup_{t \in [0, T]} \|(y_1, \dots, y_n)(t)\|_{(H_a^1)^n}^2 + \int_0^T (\|(y_{1t}, \dots, y_{nt})\|_{L^2}^2 + \|((ay_{1x})_x, \dots, (ay_{nx})_x)\|_{L^2}^2) dt \\ \leq C_T \left(\|(y_1^0, \dots, y_n^0)\|_{(H_a^1)^n}^2 + \|v\|_{(L^2(Q))^m}^2 \right) \quad (13) \end{aligned}$$

for a constant $C_T > 0$.

3. Carleman estimates for one equation. In order to establish a Carleman estimate for the adjoint system (8), we are led to see Carleman estimates already established in the case of one single parabolic degenerate equation of order 2 in space [3, 15].

$$\begin{cases} u_t - (a(x)u_x)_x + bu = f, & \text{in } Q, \\ \mathbf{C}u = 0 & \text{on } \Sigma, \\ u(T, x) = u_0(x) & \text{in } (0, 1). \end{cases} \quad (14)$$

For this purpose, let us consider the following time and space weight functions

$$\begin{cases} \theta(t) = \frac{1}{t^4(T-t)^4}, \psi(x) = \lambda \left(\int_0^x \frac{y}{a(y)} dy - c \right) \text{ and } \varphi(t, x) = \theta(t)\psi(x), \\ \Phi(t, x) = \theta(t)\Psi(x) \text{ and } \Psi(x) = e^{\rho\sigma(x)} - e^{2\rho\|\sigma\|_\infty}, \end{cases} \quad (15)$$

where σ is a $\mathcal{C}^2([0, 1])$ function such that $\sigma(x) > 0$ in $(0, 1)$, $\sigma(0) = \sigma(1) = 0$ and $\sigma_x(x) \neq 0$ in $[0, 1] \setminus \omega_0$, ω_0 is an open subset of ω , and the parameters c , ρ and λ are chosen as in [15] such that

$$c > 4^n c_0, \quad \rho > \frac{\ln\left(\frac{4^n(c-c_0)}{c-4^n c_0}\right)}{\|\sigma\|_\infty}, \quad (16)$$

$$\frac{e^{2\rho\|\sigma\|_\infty}}{c - c_0} < \lambda < \frac{4^n}{(4^n - 1)c} \left(e^{2\rho\|\sigma\|_\infty} - e^{\rho\|\sigma\|_\infty} \right), \quad (17)$$

where $c_0 = \int_0^1 \frac{x}{a(x)} dx$.

We recall first the following useful properties of the weighted functions for which the proofs are given in [15].

Lemma 3.1. *Under assumptions (16)-(17), we have*

$$\frac{4^n}{4^n - 1} \Phi < \varphi \leq \Phi, \quad (18)$$

and

$$\frac{4}{3} \Phi < \frac{4^2}{4^2 - 1} \Phi < \dots < \frac{4^{(n-1)}}{4^{(n-1)} - 1} \Phi < \frac{4^n}{4^n - 1} \Phi. \quad (19)$$

Let ω' a subset of ω and set $\omega'' := (x_1'', x_2'') \subset \subset \omega'$ and $\xi \in C^\infty([0, 1])$ such that $0 \leq \xi(x) \leq 1$ for $x \in (0, 1)$, $\xi(x) = 1$ for $x \in (0, x_1'')$ and $\xi(x) = 0$ for $x \in (x_2'', 1)$.

The two following results have been proved in [3].

Proposition 1. *Let $T > 0$ and $\tau \in \mathbb{R}$. Then there exists two positive constants C and s_0 such that, for all $u_0 \in L^2(0, 1)$, the solution u of equation (14) satisfies*

$$\begin{aligned} & \iint_Q \left((s\theta)^{\tau-1} \xi^2 u_t^2 + (s\theta)^{\tau-1} \xi^2 (\mathcal{M}u)^2 + (s\theta)^{1+\tau} a \xi^2 u_x^2 + (s\theta)^{3+\tau} \frac{x^2}{a} \xi^2 u^2 \right) e^{2s\varphi(t,x)} dx dt \\ & \leq C \left(\iint_Q \xi^2 (s\theta)^\tau f^2(t, x) e^{2s\varphi(t,x)} dx dt + \iint_{Q_{\omega'}} (s\theta)^{2+\tau} u^2 e^{2s\varphi(t,x)} dx dt \right) \end{aligned} \quad (20)$$

for all $s \geq s_0$, with $Q_{\omega'} = (0, T) \times \omega'$.

Proposition 1 gives a Carleman estimate in $(0, x_1')$. The following proposition is a non degenerate Carleman estimate to the equation (14) on the interval $(x_1', 1)$.

Proposition 2. *Let $T > 0$ and $\tau \in \mathbb{R}$. Then, there exist two positive constants C and s_0 such that for every $u_0 \in L^2(0, 1)$, the solution u of equation (14) satisfies*

$$\begin{aligned} & \iint_Q \left((s\theta)^{\tau-1} \varsigma^2 u_t^2 + (s\theta)^{\tau-1} \varsigma^2 (\mathcal{M}u)^2 + (s\theta)^{1+\tau} a \varsigma^2 u_x^2 + (s\theta)^{3+\tau} \frac{x^2}{a} \varsigma^2 u^2 \right) e^{2s\Phi} dx dt \\ & \leq C \left(\iint_Q \varsigma^2 (s\theta)^\tau f^2(t, x) e^{2s\Phi} dx dt + \iint_{Q_{\omega'}} (s\theta)^{3+\tau} u^2 e^{2s\Phi} dx dt \right) \end{aligned} \quad (21)$$

for all $s \geq s_0$, with $\varsigma = 1 - \xi$ and $Q_{\omega'} = (0, T) \times \omega'$.

4. Some useful results. Since the number of control forces is less than the number of equations, we need to highlight the equation coupling tools. Indeed, the equations are coupled by means of this algebraic condition $\text{rank}[\lambda_i \mathbf{D} - A|B] = n \ \forall i \geq 1$. Let us introduce the following operators $\mathcal{K} : D(\mathcal{K}) \subset L^2([0, 1]; \mathbb{R}^{nm}) \mapsto L^2([0, 1]; \mathbb{R}^n)$ and $\mathcal{K}_* : D(\mathcal{K}_*) \subset L^2([0, 1]; \mathbb{R}^n) \mapsto L^2([0, 1]; \mathbb{R}^{nm})$ with

$$\begin{aligned} D(\mathcal{K}) &= \{v \in L^2([0, 1]; \mathbb{R}^{nm}) : [-\mathbf{D}\mathcal{M} - A|B]v \in L^2([0, 1]; \mathbb{R}^n)\} \\ D(\mathcal{K}_*) &= \{\varphi \in L^2([0, 1]; \mathbb{R}^n) : [-\mathbf{D}\mathcal{M} - A|B]^* \varphi \in L^2([0, 1]; \mathbb{R}^{nm})\} \end{aligned}$$

and

$$\mathcal{K}v := [-\mathbf{D}\mathcal{M} - A|B]v, \quad \mathcal{K}_*\varphi := [-\mathbf{D}\mathcal{M} - A|B]^* \varphi \quad (22)$$

$\mathcal{K}$ and $\mathcal{K}_*$ are densely defined unbounded operators. we have the following estimate

Proposition 3. *Let us assume that $\text{rank}[\lambda_i \mathbf{D} - A|B] = n \ \forall i \geq 1$, then for every φ such that $(-\mathcal{M})^k(\mathcal{K}_*\varphi)$ in $L^2(0, 1)$ we have*

$$\int_0^1 |\varphi(x, t)|^2 dx \leq R \int_0^1 |(-\mathcal{M})^k(\mathcal{K}_*\varphi)(x, t)|^2 dx \quad (23)$$

for any $t \in [0, T)$ and any $k \geq (n-1)^2$, where R depends only on n , $\mathbf{D}$ and A .

Proof. We adapt the same argument as in [16] to our degenerate case. Let denote by K_i the matrices $K_i = [\lambda_i \mathbf{D} - A|B] \in \mathcal{L}(\mathbb{R}^{nm}; \mathbb{R}^n) \quad \forall i \geq 1$. Let $f \in L^2([0, 1]; \mathbb{R}^n)$ be given

$$f = \sum_{i=1}^p f_i \mathbf{w}_i \quad (24)$$

where $f_i \in \mathbb{R}^n$ for some $p \geq 1$, then

$$\mathcal{K}_* f = \sum_{i=1}^p (K_i^* f_i) \mathbf{w}_i, \quad (-\mathcal{M})^k \mathcal{K}_* f = \sum_{i=1}^p \lambda_i^k (K_i^* f_i) \mathbf{w}_i,$$

hence

$$\|(-\mathcal{M})^k \mathcal{K}_* f\|_{L^2}^2 = \sum_{i=1}^p \lambda_i^{2k} |K_i^* f_i|^2. \quad (25)$$

Let us denote by η_j^i , for $1 \leq j \leq n$, the real and nonnegative eigenvalues of $K_i K_i^*$. Then we have

$$|K_i^* f_i|^2 = (K_i K_i^* f_i, f_i) \geq \eta_1^i |f_i|^2. \quad (26)$$

There exists c_1 such that

$$\det K_i K_i^* \geq c_1 \quad \forall i \geq 1.$$

Indeed, let us set $p(\lambda) = \det \tilde{K}(\lambda) \tilde{K}(\lambda)^*$ for all λ , with

$$\tilde{K}(\lambda) := [\lambda \mathbf{D} - A|B].$$

Thus, $p(\lambda)$ is a polynomial function of degree $2n(n-1)$, $p(\lambda) \geq 0$ for all λ and $p(\lambda_i) \neq 0$ for all i . Since the roots of $p(\lambda) = 0$ are in a disk of radius R for some $R > 0$, then, there exists $C_2 > 0$ such that $p(\lambda) \geq C_2$ for $|\lambda| \geq R$. Moreover, for some ℓ , one has $\lambda_\ell > R$. Hence,

- Either $i \leq \ell - 1$ and then $\det K_i K_i^* \geq C_3 := \min_{j \leq \ell-1} \det K_j K_j^*$.
- Or $i \geq \ell$ and then $\lambda_i \geq \lambda_\ell > R$ and $\det K_i K_i^* \geq C_2$.

Thus, $\det K_i K_i^* \geq c_1 = \min(C_2, C_3)$.

Furthermore, for each $i \geq 1$ and each $\ell = 1, \dots, n$ there exists $\tilde{f}_\ell \in \mathbb{R}^n \setminus \{0\}$ such that

$$\eta_\ell^i = \frac{(K_i K_i^* \tilde{f}_\ell, \tilde{f}_\ell)}{|\tilde{f}_\ell|^2} \leq \|K_i K_i^*\|_2 \leq C_4 (1 + \lambda_i^{2(n-1)}), \quad (27)$$

where $\|\cdot\|_2$ in the usual Euclidean norm in $\mathcal{L}(\mathbb{R}^n)$. Then we infer

$$\eta_1^i = \frac{\det K_i K_i^*}{\prod_{\ell \geq 2} \eta_\ell^i} \geq \frac{c_1}{C_4^{n-1} (1 + \lambda_i^{2(n-1)})^{n-1}} \geq C_5 \lambda_i^{-2(n-1)^2}.$$

Coming back to (26) we get

$$|K_i^* f_i|^2 = (K_i K_i^* f_i, f_i) \geq \eta_1^i |f_i|^2 \geq C_5 \lambda_i^{-2(n-1)^2} |f_i|^2.$$

Therefore

$$\|(-\mathcal{M})^k \mathcal{K}_* f\|_{L^2}^2 = \sum_{i=1}^p \lambda_i^{2k} |K_i^* f_i|^2 \geq C_5 \sum_{i=1}^p \lambda_i^{2(k-(n-1)^2)} |f_i|^2 \geq C \|f\|_{L^2}^2.$$

As this is true for all f spanned by a finite amount of the $\mathbf{w}_i$, then we infer that this must also hold for all $f \in L^2((0, 1); \mathbb{R}^n)$ such that $(-\mathcal{M})^k \mathcal{K}_* f \in L^2((0, 1); \mathbb{R}^n)$. In particular, we obtain the estimate (23). $\square$

From now on, we consider ϕ with the monomial derivative $\mathcal{M}^i \partial_t^j \phi \in L^2(0, T; H_a^2(0, 1))$ for every $i, j \in \mathbb{N}$, a solution of the following scalar degenerate parabolic equation of order $2n$ in space

$$\begin{cases} P(\partial_t, \mathcal{M})\phi = 0 & \text{in } Q, \\ \mathbf{C}\mathcal{M}^k \phi = 0, \quad k \geq 0, & \text{on } \Sigma, \end{cases} \quad (28)$$

where $P(\partial_t, \mathcal{M})$ is the operator defined by $P(\partial_t, \mathcal{M}) = \det(\partial_t I_d + \mathbf{D}^* \mathcal{M} + A^*)$.

In fact, we prove that all components of every solution of the adjoint system (8) are solutions of the scalar PDE (28). So, it will be necessary to establish Carleman estimate for the scalar PDE (28). We recall first the following result given in [3, 7, 16].

Proposition 4. *Let $z_0 \in \mathbb{D}^n$ and let $z = (z_1, \dots, z_n)^*$ be the corresponding solution of problem (8). Then, $z \in \mathcal{C}^k([0, T]; D(\mathcal{M}^p)^n)$ for every $k, p \geq 0$, and for every i , z_i solves equation (28).*

The following proposition is the crucial result in this paper, since it generalize [16, Lemma 4.1] to the case where the Jordan block size exceeds 4.

Proposition 5. *for any $k \geq 0$ and $j \geq 0$, we can find an integer $m(k, j) \geq 0$, a constant $C(k, j) > 0$ and an open set $\omega(k, j)$ satisfying $\omega \subseteq \omega(k, j) \subseteq \omega_1$, such that*

$$I(\tau, (-\mathcal{M})^k \partial_t^j \phi) \leq C(k, j) \iint_{(0, T) \times \omega(k, j)} (s\theta)^{m(k, j)} |\phi|^2 e^{2s\Phi} dx dt \quad (29)$$

where ϕ satisfies (28) and

$$I(\tau, z) = \iint_Q \left((s\theta)^{\tau-1} z_t^2 + (s\theta)^{\tau-1} (\mathcal{M}z)^2 + (s\theta)^{\tau+1} a(x) z_x^2 + (s\theta)^{\tau+3} \frac{x^2}{a(x)} z^2 \right) e^{2s\varphi} dx dt.$$

Proof. We will prove (29) by induction on k and j in two steps.

Step 1. Proof of (29) for $k = j = 0$

Let us see that, for s large enough, one has

$$I(\tau, \phi) \leq C(0, 0) \iint_{(0, T) \times \omega(0, 0)} (s\theta)^{m(0, 0)} |\phi|^2 e^{2s\Phi} dx dt \quad (30)$$

for some integer $m(0, 0)$, a positive constant $C(0, 0)$ and an open subset $\omega(0, 0)$ satisfying $\omega \subseteq \omega(0, 0) \subseteq \omega_1$.

We assume that $\mathbf{D}$ satisfies the assumptions (2)-(4), then we have for some $p \geq 1$

$$I_d \partial_t + \mathbf{D}^* \mathcal{M} + A^* = \begin{bmatrix} H_1(\partial_t, \mathcal{M}) & A_{21}^* & \cdots & A_{p1}^* \\ A_{12}^* & H_2(\partial_t, \mathcal{M}) & \cdots & A_{p2}^* \\ \vdots & \vdots & \ddots & \vdots \\ A_{1p}^* & A_{2p}^* & \cdots & H_p(\partial_t, \mathcal{M}) \end{bmatrix} \quad (31)$$

where $H_i(\partial_t, \mathcal{M})$ is the non-scalar operator $H_i(\partial_t, \mathcal{M}) := I_d \partial_t + J_i^* \mathcal{M} + A_{ii}^*$, the J_i^* are Jordan blocks, i.e. each of them is of the form (4) for some $d_i \in \mathbb{C}$ and the A_{ij} provide the corresponding block decomposition of A . Thus we can write (28) as follow

$$\prod_{i=1}^p \det H_i(\partial_t, \mathcal{M}) \phi = F(\phi) \quad (32)$$

in the term $F(\phi)$ we find the composition of at most $p - 2$ operators of kind $\det H_i(\partial_t, \mathcal{M})$ applied to ϕ . Let us define the functions ψ_i by

$$\psi_1 = \phi, \psi_2 = \det H_1(\partial_t, \mathcal{M})\psi_1, \dots, \psi_p = \det H_{p-1}(\partial_t, \mathcal{M})\psi_{p-1}.$$

Thus, the equation (32) can be written as

$$\begin{cases} \det H_p(\partial_t, \mathcal{M})\psi_p = F(\phi), \\ \det H_{p-1}(\partial_t, \mathcal{M})\psi_{p-1} = \psi_p, \\ \dots \quad \dots \quad \dots \\ \det H_1(\partial_t, \mathcal{M})\psi_1 = \psi_2. \end{cases} \quad (33)$$

By hypothesis, we have

$$\mathbf{C}\phi = \mathbf{C}\psi_2 = \dots = \mathbf{C}\psi_p = 0 \quad \text{on } \Sigma.$$

Let us consider the first PDE of (33), assume that J_p is the Jordan block of dimension r associated to the complex eigenvalue α with $\text{Re}(\alpha) > 0$ and let denote by $\eta_1, \dots, \eta_r$ the diagonal components of A_{pp} . Then this PDE can be rewritten as

$$\prod_{i=1}^r (\partial_t + \alpha\mathcal{M} + \eta_i)\psi_p = F(\phi) - G(\psi_p), \quad (34)$$

where $G(\psi_p)$ is a linear combination of partial derivatives of ψ_p .

Again, let us introduce the new variables

$$\zeta_r = \psi_p, \zeta_{r-1} = (\partial_t + \alpha\mathcal{M} + \eta_r)\zeta_r, \dots, \zeta_1 = (\partial_t + \alpha\mathcal{M} + \eta_2)\zeta_2.$$

Therefore, we can rewrite (34) as a first-order system for the ζ_i :

$$\begin{cases} (\partial_t + \alpha\mathcal{M} + \eta_1)\zeta_1 = F(\phi) - G(\psi_p), \\ (\partial_t + \alpha\mathcal{M} + \eta_2)\zeta_2 = \zeta_1, \\ \dots \quad \dots \quad \dots \\ (\partial_t + \alpha\mathcal{M} + \eta_r)\zeta_r = \zeta_{r-1}. \end{cases} \quad (35)$$

with

$$\mathbf{C}\zeta_1 = \mathbf{C}\zeta_2 = \dots = \mathbf{C}\zeta_r = 0 \quad \text{on } \Sigma.$$

Notice that $|G(\psi_p)|^2$ is bounded by a sum of squares of derivatives of ψ_p . More precisely, we have $|G(\psi_p)|^2 \leq CI_G(\psi_p)$, with

$$I_G(\psi_p) := \sum_{\ell=0}^{r-1} \sum_{j_1, \dots, j_\ell=0}^r \sum_{b=0}^{r-(\ell+1)} |(-\mathcal{M})^b \prod_{i=1}^{\ell} (\partial_t + \alpha\mathcal{M} + \eta_{j_i})\psi_p|^2. \quad (36)$$

The following $I_{\alpha, \mathbf{A}}(\tau, z)$ term already used in [3] is defined by

$$I_{\alpha, \mathbf{A}}(\tau, z) = \iint_Q \left((s\theta)^{\tau-1} \alpha^2 z_t^2 + (s\theta)^{\tau-1} \alpha^2 (\mathcal{M}z)^2 + (s\theta)^{\tau+1} a(x) \alpha^2 z_x^2 + (s\theta)^{\tau+3} \frac{x^2}{a(x)} \alpha^2 z^2 \right) e^{2s\mathbf{A}} dx dt$$

where $\mathbf{A} \in \{\varphi, \Phi\}$ and $\alpha \in \{\xi, \varsigma\}$ used in Proposition 1 and Proposition 2.

Applying Carleman estimates (20) established in Proposition 1 to the first PDE in system (35), we get

$$I_{\xi, \varphi}(\tau, \zeta_1) \leq C \left(\iint_Q \xi^2 (s\theta)^\tau (|F(\phi)|^2(t, x) + I_G(\psi_p)) e^{2s\varphi} dx dt + \iint_{Q_{\omega'}} (s\theta)^{2+\tau} \zeta_1^2 e^{2s\varphi} dx dt \right).$$

And for the j^{th} PDE in system (35) where $j = 2, \dots, r$, we have

$$I_{\xi, \varphi}(\tau, \zeta_j) \leq C \left(\iint_Q \xi^2(s\theta)^\tau |\zeta_{j-1}|^2 e^{2s\varphi} dx dt + \iint_{Q_{\omega'}} (s\theta)^{\tau+2} |\zeta_j|^2 e^{2s\varphi} dx dt \right).$$

The first term in the right hand-side can be bounded as follow

$$\begin{aligned} & \iint_Q \xi^2(s\theta)^\tau |\zeta_{j-1}|^2 e^{2s\varphi} dx dt \\ &= \iint_Q (\xi(s\theta)^{\tau/2} \sqrt{\frac{a}{x^2}} |\zeta_{j-1}| e^{s\varphi}) (\xi(s\theta)^{\tau/2} \sqrt{\frac{x^2}{a}} |\zeta_{j-1}| e^{s\varphi}) dx dt \\ &\leq \iint_Q \xi^2(s\theta)^\tau \frac{a}{x^2} |\zeta_{j-1}|^2 e^{2s\varphi} dx dt + \iint_Q \xi^2(s\theta)^\tau \frac{x^2}{a} |\zeta_{j-1}|^2 e^{2s\varphi} dx dt. \end{aligned} \quad (37)$$

Knowing that $(\xi \zeta_{j-1} e^{s\varphi})_x = \xi_x \zeta_{j-1} e^{s\varphi} + \xi \zeta_{j-1x} e^{s\varphi} + \xi s \theta \frac{x}{a} \zeta_{j-1} e^{s\varphi}$ and using Hardy-Poincaré inequality and $\text{supp} \xi_x \subset Q_{\omega'}$ we get

$$\begin{aligned} & \iint_Q \xi^2(s\theta)^\tau \frac{a}{x^2} |\zeta_{j-1}|^2 e^{2s\varphi} dx dt \\ &= \iint_Q (s\theta)^\tau \frac{a}{x^2} (\xi \zeta_{j-1} e^{s\varphi})^2 dx dt \\ &\leq C \iint_Q (s\theta)^\tau a (\xi \zeta_{j-1} e^{s\varphi})_x^2 dx dt \\ &\leq C \iint_Q (s\theta)^\tau a (\xi_x \zeta_{j-1} e^{s\varphi} + \xi \zeta_{j-1x} e^{s\varphi} + \xi s \theta \frac{x}{a} \zeta_{j-1} e^{s\varphi})^2 dx dt \\ &\leq C \iint_{Q_{\omega'}} (s\theta)^\tau \zeta_{j-1}^2 e^{2s\varphi} dx dt + C \iint_Q (s\theta)^\tau \xi^2 a \zeta_{j-1x}^2 e^{2s\varphi} dx dt \\ &\quad + C \iint_Q (s\theta)^{\tau+2} \xi^2 \frac{x^2}{a} \zeta_{j-1}^2 e^{2s\varphi} dx dt. \end{aligned} \quad (38)$$

Consequently, an appropriate linear combination of the terms in the left hand sides absorbs the global weighted integrals of $|\zeta_j|^2$ for $j = 2, \dots, r$.

$$\begin{aligned} \sum_{j=1}^r I_{\xi, \varphi}(\tau, \zeta_j) &\leq C \left(\sum_{j=1}^r \iint_{Q_{\omega'}} (s\theta)^{\tau+2} |\zeta_j|^2 e^{2s\varphi} dx dt \right) \\ &\quad + C \left(\iint_Q \xi^2(s\theta)^\tau (|F(\phi)|^2(t, x) + I_G(\psi_p)) e^{2s\varphi} dx dt \right). \end{aligned} \quad (39)$$

Now let us apply the operators $(-\mathcal{M})^k$ to the j^{th} PDE in system (35) where $k = 1, \dots, r-1$ and $j = k+1, \dots, r$, we have

$$(\partial_t + \alpha \mathcal{M} + \eta_j)(-\mathcal{M})^k \zeta_j = (-\mathcal{M})^k \zeta_{j-1}.$$

By induction and using Proposition 1, we prove the following estimates

$$I_{\xi, \varphi}(\tau, (-\mathcal{M})^k \zeta_j) \leq C \left(\sum_{i=0}^k I_{\xi, \varphi}(\tau - 2i, \zeta_{j-i}) \right). \quad (40)$$

Terms in the right-hand side can be bounded by $\sum_{i=0}^k I_{\xi,\varphi}(\tau, \zeta_{j-i})$ for s enough large.

Let us denote by $\mathcal{J}_{\xi,\varphi}(\tau, \zeta)$ the following sum

$$\mathcal{J}_{\xi,\varphi}(\tau, \zeta) = \sum_{k=0}^{r-1} \sum_{j=k+1}^r I_{\xi,\varphi}(\tau, (-\mathcal{M})^k \zeta_j) \quad (41)$$

where $\zeta = (\zeta_1, \dots, \zeta_r)$. Then, we can add all these new terms (40) to the left hand side of (39) and obtain

$$\begin{aligned} \mathcal{J}_{\xi,\varphi}(\tau, \zeta) \leq & C \left(\sum_{j=1}^r \iint_{Q_{\omega'}} (s\theta)^{\tau+2} |\zeta_j|^2 e^{2s\varphi} dx dt \right) \\ & + C \left(\iint_Q \xi^2 (s\theta)^\tau (|F(\phi)|^2(t, x) + I_G(\psi_p)) e^{2s\varphi} dx dt \right) \end{aligned} \quad (42)$$

where C is a new positive constant, and s sufficiently large. From now on, we fix s sufficiently large and we try to replace the local terms in (42) corresponding to $\zeta_1, \dots, \zeta_{r-1}$ by a term of the form $(\psi_p = \zeta_r)$ using the same computation [15, Lemma 3.7], we can show the existence of a constant $C > 0$ and an integer ℓ_1 such that :

$$\mathcal{J}_{\xi,\varphi}(\tau, \zeta) \leq C \iint_{Q_{\omega'}} s^{\ell_1} \theta^{\ell_1} \psi_p^2 e^{2s\varphi} dx dt + C \iint_Q \xi^2 s^\tau \theta^\tau (|F(\phi)|^2 + I_G(\psi_p)) e^{2s\varphi} dx dt. \quad (43)$$

Since the operators $(\partial_t + \alpha\mathcal{M} + \eta_j)$, $j = 1, \dots, r$, commute, then the formula (34) can be rewritten equivalently in the form

$$\prod_{i=1}^r (\partial_t + \alpha\mathcal{M} + \eta_{\sigma(i)}) \psi_p = F(\phi) - G(\psi_p),$$

where σ is any permutation in $\mathcal{P}_r$. Hence, we can introduce the new variables

$$\zeta_r^\sigma = \psi_p, \zeta_{r-1}^\sigma = (\partial_t + \alpha\mathcal{M} + \eta_{\sigma(r)}) \zeta_r^\sigma, \dots, \zeta_1^\sigma = (\partial_t + \alpha\mathcal{M} + \eta_{\sigma(2)}) \zeta_2^\sigma$$

and we can also rewrite (34) as a first-order system for the ζ_i^σ :

$$\begin{cases} (\partial_t + \alpha\mathcal{M} + \eta_{\sigma(1)}) \zeta_1^\sigma = F(\phi) - G(\psi_p), \\ (\partial_t + \alpha\mathcal{M} + \eta_{\sigma(2)}) \zeta_2^\sigma = \zeta_1^\sigma, \\ \dots \quad \dots \quad \dots \\ (\partial_t + \alpha\mathcal{M} + \eta_{\sigma(r)}) \zeta_r^\sigma = \zeta_{r-1}^\sigma. \end{cases} \quad (44)$$

Again, with

$$\mathbf{C} \zeta_1^\sigma = \mathbf{C} \zeta_2^\sigma = \dots = \mathbf{C} \zeta_r^\sigma = 0 \quad \text{on } \Sigma.$$

Similarly, we obtain an estimate like (43)

$$\mathcal{I}_{\xi,\varphi}(\tau, \zeta^\sigma) \leq C \iint_{Q_{\omega'}} s^{\ell_1} \theta^{\ell_1} \psi_p^2 e^{2s\varphi} dx dt + C \iint_Q \xi^2 s^\tau \theta^\tau (|F(\phi)|^2 + I_G(\psi_p)) e^{2s\varphi} dx dt.$$

In this inequality we have used the notation $\zeta^\sigma = (\zeta_1^\sigma, \dots, \zeta_r^\sigma)$. Now, let us define $\mathcal{I}_{\xi,\varphi}(\tau, \zeta)$ by

$$\mathcal{I}_{\xi,\varphi}(\tau, \psi_p) = \sum_{\sigma \in \mathcal{P}_r} \mathcal{J}_{\xi,\varphi}(\tau, \zeta^\sigma).$$

We have

$$\mathcal{I}_{\xi,\varphi}(\tau, \psi_p) \leq C \iint_{Q_{\omega'}} s^{\ell_1} \theta^{\ell_1} \psi_p^2 e^{2s\varphi} dx dt + C \iint_Q \xi^2 s^\tau \theta^\tau (|F(\phi)|^2 + I_G(\psi_p)) e^{2s\varphi} dx dt.$$

Observe that all the terms in $I_G(\psi_p)$ are also in the left multiplied by weights of the form $s^\tau \theta^\tau e^{2s\varphi}$. Consequently, for sufficiently large s , these terms are absorbed and we find

$$\mathcal{I}_{\xi,\varphi}(\tau, \psi_p) \leq C \iint_{Q_{\omega'}} s^{\ell_1} \theta^{\ell_1} \psi_p^2 e^{2s\varphi} dxdt + C \iint_Q \xi^2 s^\tau \theta^\tau |F(\phi)|^2 e^{2s\varphi} dxdt. \quad (45)$$

Let us now consider the second PDE in (33). Arguing in the same way, we deduce the following estimate for ψ_{p-1} :

$$\mathcal{I}_{\xi,\varphi}(\tau, \psi_{p-1}) \leq C \iint_{Q_{\omega'}} s^{\ell_2} \theta^{\ell_2} |\psi_{p-1}|^2 e^{2s\varphi} dxdt + C \iint_Q \xi^2 s^\tau \theta^\tau |\psi_p|^2 e^{2s\varphi} dxdt. \quad (46)$$

The corresponding similar estimate also holds for $\psi_{p-2}, \dots, \psi_1$. Thus, adding the previous terms and taking into account that $\psi_1 = \phi$ and the global integrals of $\psi_p, \dots, \psi_2$ in the right hand side are smaller than the terms in the left, we get an estimate for all the ψ_i :

$$\begin{aligned} \sum_{i=1}^p \mathcal{I}_{\xi,\varphi}(\tau, \psi_i) &\leq C \left(\iint_Q \xi^2 s^\tau \theta^\tau |F(\phi)|^2 e^{2s\varphi} dxdt \right. \\ &\quad \left. + \iint_{Q_{\omega'}} (s\theta)^{\ell_2} |\phi|^2 e^{2s\varphi} dxdt + \sum_{i=2}^p \iint_{Q_{\omega'}} (s\theta)^{\ell_2} |\psi_i|^2 e^{2s\varphi} dxdt \right). \end{aligned} \quad (47)$$

Again, using the cascade structure of system (33), all the local integrals in the right can be absorbed by the left hand side, with the exception of the local weighted integral of $|\phi|^2$. All we have to do is to enlarge the open set ω' and argue like in the passage from (42) to (43). Therefore, the following is obtained:

$$\sum_{i=1}^p \mathcal{I}_{\xi,\varphi}(\tau, \psi_i) \leq C \left(\iint_Q \xi^2 s^\tau \theta^\tau |F(\phi)|^2 e^{2s\varphi} dxdt + \iint_{Q_{\omega'}} (s\theta)^{\ell_2} |\phi|^2 e^{2s\varphi} dxdt \right). \quad (48)$$

Taking into account that the operators $\det H_i(\partial_t, \mathcal{M})$, $i = 1, \dots, p$, commute we can rewrite (32) in the form

$$\prod_{i=1}^p \det H_{\sigma(i)}(\partial_t, \mathcal{M}) \phi = F(\phi)$$

where σ is any permutation in $\mathcal{P}_n$. This means that another equivalent formulation of (28) is

$$\begin{cases} \det H_{\sigma(p)}(\partial_t, \mathcal{M}) \psi_p^\sigma = F(\phi), \\ \det H_{\sigma(p-1)}(\partial_t, \mathcal{M}) \psi_{p-1}^\sigma = \psi_p^\sigma, \\ \dots \quad \dots \quad \dots \\ \det H_{\sigma(1)}(\partial_t, \mathcal{M}) \psi_1^\sigma = \psi_2^\sigma \end{cases}$$

and we obtain an estimate like (48) where the terms in the left hand sides are a global weighted integrals of $\phi, \psi_2^\sigma, \dots, \psi_{p-1}^\sigma$ and ψ_p^σ and in the right hand sides we have terms concerning $F(\phi)$ and ϕ . Recall that $F(\phi)$ is a sum of terms where, at most, $p-2$ operators of the kind $\det H_j(\partial_t, \mathcal{M})$ are applied to ϕ . Since σ is arbitrary in $\mathcal{P}_n$, using all these possible estimates together and arguing as above, it becomes also clear that the terms containing $|F(\phi)|^2$ can be absorbed by the terms in the left. This gives

$$\sum_{\sigma \in \mathcal{P}_n} \sum_{i=1}^p \mathcal{I}_{\xi,\varphi}(\tau, \psi_i^\sigma) \leq C \iint_{Q_{\omega'}} (s\theta)^{\ell_2} |\phi|^2 e^{2s\varphi} dxdt. \quad (49)$$

Likewise, by applying Proposition 2 we infer

$$\sum_{\sigma \in \mathcal{P}_n} \sum_{i=1}^p \mathcal{I}_{\varsigma, \Phi}(\tau, \psi_i^\sigma) \leq C \iint_{Q_{\omega'}} (s\theta)^{\ell_3} |\phi|^2 e^{2s\Phi} dxdt. \quad (50)$$

From (49) and (51) we deduce

$$\sum_{\sigma \in \mathcal{P}_n} \sum_{i=1}^p \mathcal{I}(\tau, \psi_i^\sigma) \leq C \iint_{Q_{\omega'}} (s\theta)^{\tilde{\ell}} |\phi|^2 e^{2s\Phi} dxdt. \quad (51)$$

where $\tilde{\ell} = \max(\ell_2, \ell_3)$ and $\mathcal{I}(\tau, \psi_i^\sigma) = \mathcal{I}_{\xi, \varphi}(\tau, \psi_i^\sigma) + \mathcal{I}_{\varsigma, \Phi}(\tau, \psi_i^\sigma)$. This proves (30).

Step 2. Induction on k and j .

Assume that (29) is true for any $k' = 0, 1, \dots, k$ and $j' = 0, 1, \dots, j$ and any ϕ solution to (28) satisfying the assumptions of the Proposition 5. Let us prove (29) (for instance) with k replaced by $k+1$; the proof with the same k and j replaced by $j+1$ is essentially the same. Since $\hat{\phi} := (-\mathcal{M})\phi$ also satisfies (28), we have by hypothesis

$$\begin{aligned} I(\tau, (-\mathcal{M})^{k+1} \partial_t^j \phi) &= I(\tau, (-\mathcal{M})^k \partial_t^j \hat{\phi}) \\ &\leq C(k, j) \iint_{(0, T) \times \omega(k, j)} (s\theta)^{m(k, j)} |\hat{\phi}|^2 e^{2s\Phi} dxdt \\ &\leq C(k, j) \iint_{(0, T) \times \omega(k, j)} (s\theta)^{m(k, j)} |\mathcal{M}\phi|^2 e^{2s\Phi} dxdt \\ &\leq C(k, j) C' \iint_{(0, T) \times \omega(k, j)} (s\theta)^{m(k, j)} |\mathcal{M}\phi|^2 e^{2s\varphi} dxdt \\ &\leq C(k, j) C' I(m(k, j) + 1, \phi). \end{aligned}$$

Applying the formula (30), there exist a positive constant C'' an integer m' and an open subset $\mathcal{O}'$ satisfying $\omega \Subset \mathcal{O}' \Subset \omega_1$ such that

$$I(m(k, j) + 1, \phi) \leq C'' \iint_{(0, T) \times \mathcal{O}'} (s\theta)^{m'} |\phi|^2 e^{2s\Phi} dxdt.$$

Thus, we have

$$I(\tau, (-\mathcal{M})^{k+1} \partial_t^j \phi) \leq C(k+1, j) \iint_{(0, T) \times \omega(k+1, j)} (s\theta)^{m(k+1, j)} |\phi|^2 e^{2s\Phi} dxdt$$

with $m(k+1, j) = m'$, $C(k+1, j) = C(k, j) C' C''$ and $\omega(k+1, j) = \mathcal{O}'$. This ends the proof. $\square$

5. Proof of the main result.

Proof of Theorem 1.1. Let us first assume that the system (1) is null-controllable. If we have $\text{rank}[\lambda_i \mathbf{D} + A|B] \leq n-1$ for some i , then the associated ordinary differential system is not null-controllable. This means there exists $y_T \in \mathbb{R}^n \setminus \{0\}$ such that the solution to the Cauchy problem

$$\begin{cases} -\partial_t y + (\lambda_i \mathbf{D}^* + A^*)y = 0 & \text{in } Q, \\ y(T) = y_T & \text{on } (0, T), \end{cases} \quad (52)$$

satisfies

$$B^* y(t) = 0.$$

Taking $z_T = y_T \mathbf{w}_i$ in the adjoint system (8), where $\mathbf{w}_i$ is an eigenfunction associated to λ_i , we can see as in [3] that the corresponding solution cannot satisfy the observability inequality (9). Consequently, the condition (10) must hold.

Conversely, let us assume that (10) is satisfied, and let us prove that the system (1) is null-controllable.

Let z be the solution to the adjoint system (8) corresponding to a final data z_T , by Proposition 3, for $k \geq (n-1)^2$ there exists a positive constant C depends on n , $\mathbf{D}$ and A such that

$$\int_0^1 |z(x, t)|^2 dx \leq C \int_0^1 |(-\mathcal{M})^k (K_* z)(x, t)|^2 dx \quad (53)$$

for any $t \in [0, T]$. From (22) the components of $K_* z$ are appropriate linear combinations of the components of z and their second-order in space derivatives. Notice again that, for all $t \in [0, T]$, $z(\cdot, t)$ is regular enough to give a sense to $(-\mathcal{M})^k (K_* z)$, which belongs to $L^2(0, 1)$.

By Proposition 4 $z \in \mathcal{C}^k([0, T]; D(\mathcal{M}^p)^n)$ for every $k, p \geq 0$ and for every i the component z_i of z solves equation (28). Thus, we can write (29) for any component of $B^* z$. This gives the following inequality for all $j, k \geq 0$ and all $\ell = 1, \dots, m$

$$\begin{aligned} \iint_Q |(B^*((-\mathcal{M})^k \partial_t^j z))_\ell|^2 e^{2s\varphi} dx dt &= \iint_Q |(-\mathcal{M})^k \partial_t^j (B^* z)_\ell|^2 e^{2s\varphi} dx dt \\ &\leq C(k, j) \iint_{(0, T) \times \omega(k, j)} |(B^* z)_\ell|^2 e^{2s\Phi} dx dt. \end{aligned}$$

Let $M_0 = \max_{x \in (0, 1)} |\psi(x)|$, thus

$$\begin{aligned} \iint_Q |(-\mathcal{M})^k K_* z|^2 e^{-2s\theta M_0} dx dt &\leq C \sum_{\ell=1}^m \iint_Q |(B^*((-\mathcal{M})^k \partial_t^j z))_\ell|^2 e^{2s\varphi} dx dt \\ &\leq C \sum_{\ell=1}^m C(k, j) \iint_{(0, T) \times \omega} |(B^* z)_\ell|^2 e^{2s\Phi} dx dt \\ &\leq C \iint_{(0, T) \times \omega} |B^* z|^2 e^{2s\Phi} dx dt. \end{aligned} \quad (54)$$

From the estimates (53) and (54), we will easily deduce the observability inequality (9) and, therefore, the null controllability of system (1). $\square$

6. Null controllability for semilinear systems. Now we consider the following semi-linear non-diagonalizable parabolic degenerate systems.

$$\begin{cases} \partial_t Y = \mathbf{D} \mathcal{M} Y + F(Y) + B v \mathbb{1}_\omega & \text{in } Q, \\ \mathbf{C} Y = 0 & \text{on } \Sigma, \\ Y(0, x) = Y_0(x) & \text{in } (0, 1), \end{cases} \quad (55)$$

where $F \in \mathcal{C}^1(\mathbb{R}^n)$ is a globally Lipschitz function depending only on Y and $F(0) = 0$. Our goal is to study the null controllability of the system (55). To this end, we adapt a standard strategy, as in [24, 8, 10, 1], which consists in using the linearization technique, the approximate null controllability, the variational approach and the Schauder fixed point theorem. The system (55) can be written as follow

$$\begin{cases} \partial_t Y = (\mathbf{D} \mathcal{M} + A_Y) Y + B v \mathbb{1}_\omega & \text{in } Q, \\ \mathbf{C} Y = 0 & \text{on } \Sigma, \\ Y(0, x) = Y_0(x) & \text{in } (0, 1), \end{cases}$$

where A_Y is the matrix defined by $a_{i,j}^Y = \int_0^1 \partial_j F_i(\tau Y) d\tau$.

In the sequel, we assume

$$\text{rank}[\lambda_i \mathbf{D} - A_Y | B] = n \quad \forall i \geq 1 \quad \text{for all } Y \in (L^2(0,1))^n. \quad (56)$$

Let us recall the set X_T (see Theorem 2.2) induced with the norm

$$\|Y\|_{X_T}^2 = \sup_{t \in [0,T]} \|Y(t)\|_{(L^2(0,1))^n}^2 + \int_0^T \|\sqrt{a}Y(t)\|_{(L^2(0,1))^n}^2 dt.$$

For a fixed $\tilde{Y}$ in X_T , consider the associated linear system

$$\begin{cases} \partial_t Y = (\mathbf{D}\mathcal{M} + A_{\tilde{Y}})Y + Bv\mathbb{1}_\omega & \text{in } Q, \\ \mathbf{C}Y = 0 & \text{on } \Sigma, \\ Y(0, x) = Y_0(x) & \text{in } (0, 1), \end{cases} \quad (57)$$

and its adjoint system

$$\begin{cases} \partial_t Z = (\mathbf{D}^*\mathcal{M} + A_{\tilde{Y}}^*)Z & \text{in } Q, \\ \mathbf{C}Z = 0 & \text{on } \Sigma, \\ Z(0, x) = Z_0(x) & \text{in } (0, 1). \end{cases} \quad (58)$$

Thus, from the assumption (56), it follows that the pair of matrices $(A_{\tilde{Y}}, B)$ satisfies the algebraic condition (10), which means that the linearized system (57) is null controllable. In order to construct a suitable fixed point operator, we start at first by proving the uniqueness of the control with minimal norm. For a given $\varepsilon > 0$ and $Y_0 \in (L^2(0,1))^n$ we consider the following functional

$$\begin{aligned} J_{\varepsilon, \tilde{Y}}(v) &= \frac{1}{2} \int_0^T \|v\|_{(L^2(0,1))^n}^2 dt + \frac{1}{2\varepsilon} \|Y(T)\|_{(L^2(0,1))^n}^2, \\ J_{\varepsilon, \tilde{Y}}^*(Z_0) &= \frac{1}{2} \iint_{Q_\omega} |B^*Z|^2 + \frac{\varepsilon}{2} \|Z_0\|_{(L^2(0,1))^n}^2 + \int_0^1 \left(\sum_{i=1}^n Z_i(T) Y_{i0} \right) dx \end{aligned} \quad (59)$$

where Y is the solution of system (57) with initial data Y_0 and Z is the solution of (58) with initial data Z_0 . By a classical arguments, minimization problems

$$\min\{J_{\varepsilon, \tilde{Y}}(v), Bv \in L^2(Q)^n\} \quad \text{and} \quad \min\{J_{\varepsilon, \tilde{Y}}^*(Z_0), Z_0 \in (L^2(0,1))^n\}$$

admit unique solutions $v^{\varepsilon, \tilde{Y}}$ and $Z_0^{\varepsilon, \tilde{Y}}$ such that

$$v = B^*Z\mathbb{1}_\omega, \quad (60)$$

$$Z_0^{\varepsilon, \tilde{Y}} = -\frac{1}{\varepsilon} Y^{\varepsilon, \tilde{Y}}(T) \quad (61)$$

where $Y^{\varepsilon, \tilde{Y}}$ is the solution of (57) associated to the control v and $Z^{\varepsilon, \tilde{Y}}$ is the solution of the adjoint problem (58) with the initial data $Z_0^{\varepsilon, \tilde{Y}}$. Since $J_{\varepsilon, \tilde{Y}}^*(Z_0^{\varepsilon, \tilde{Y}}) \leq 0$, then from (59) and (61) we infer

$$\frac{1}{2} \iint_{Q_\omega} |B^*Z|^2 + \frac{1}{2\varepsilon} \|Y^{\varepsilon, \tilde{Y}}(T)\|_{(L^2(0,1))^n}^2 \leq \|Z(T, \cdot)\|_{(L^2(0,1))^n} \|Y(0, \cdot)\|_{(L^2(0,1))^n}. \quad (62)$$

On the other hand, by the observability inequality (9)

$$\|Z(T, \cdot)\|_{(L^2(0,1))^n}^2 \leq C \iint_{Q_\omega} |B^*Z(t, x)|^2. \quad (63)$$

From (60) and the estimates (62) – (63) we infer

$$\frac{1}{2} \int_0^T \|v^{\varepsilon, \tilde{Y}}\|_{(L^2(0,1))^m}^2 dt + \frac{1}{2\varepsilon} \|Y^{\varepsilon, \tilde{Y}}(T)\|_{(L^2(0,1))^n}^2 \leq C \|Y(0, \cdot)\|_{(L^2(0,1))^n}^2. \quad (64)$$

The uniqueness of the control $v^{\varepsilon, \tilde{Y}}$ allows to define the operator

$$\begin{aligned} K_\varepsilon &: X_T \rightarrow X_T \\ \tilde{Y} &\mapsto Y^{\varepsilon, \tilde{Y}}. \end{aligned}$$

Any fixed point Y^ε of K_ε is a solution of the semilinear system (55) associated to $v^{\varepsilon, Y^\varepsilon}$ and it satisfies

$$\|Y^\varepsilon\|_{(L^2(0,1))^n}^2 \leq \varepsilon C.$$

Indeed, suppose first that $Y_0 \in (H_a^1)^n$. From (13) and since the matrix A is constant, we have the following estimate

$$\begin{aligned} \sup_{t \in [0, T]} \|Y^{\varepsilon, \tilde{Y}}(t)\|_{(H_a^1)^n}^2 + \int_0^T \left(\|\partial_t Y^{\varepsilon, \tilde{Y}}\|_{L^2}^2 + \|\mathcal{M} Y^{\varepsilon, \tilde{Y}}\|_{L^2}^2 \right) dt \\ \leq C_T \left(\|Y_0^{\varepsilon, \tilde{Y}}\|_{(H_a^1)^n}^2 + \|v^{\varepsilon, \tilde{Y}}\|_{(L^2(Q))^m}^2 \right). \end{aligned} \quad (65)$$

Thus, by the estimate (64) we infer

$$\|Y^{\varepsilon, \tilde{Y}}\|_{X_T} \leq C \|Y_0\|_{(H_a^1)^n}, \quad (66)$$

$$\|Y^{\varepsilon, \tilde{Y}}\|_{Y_T} \leq C \|Y_0\|_{(H_a^1)^n} \quad (67)$$

where $Y_T := H^1(0, T; (L^2(0, 1))^n) \cap L^2(0, T; (H_a^1)^n)$ with the norm

$$\|Y\|_{Y_T} = \int_0^T \left(\|Y(t)\|_{(H_a^1)^n}^2 + \|\partial_t Y\|_{L^2}^2 + \|\mathcal{M} Y\|_{L^2}^2 \right) dt.$$

Thus, the range of K_ε is include in the ball $B(0, R)$ of X_T with the radius $R = C \|Y_0\|_{(H_a^1)^n}$ where C is the constant used in (66). Then $K_\varepsilon(B(0, R)) \subset B(0, R)$. Now let us prove that the operator K_ε is continuous and compact. The compactness of K_ε results from the compactness of the embedding

$$Y_T \hookrightarrow X_T \quad (68)$$

see [10, Theorem 4.4]. For the continuity, let us consider the sequence $\tilde{Y}_n$ that converge to $\tilde{Y}$ in X_T . To simplify, let denote $Y^{\varepsilon, \tilde{Y}_n}$ and $v^{\varepsilon, \tilde{Y}_n}$ respectively by Y_n and v_n (for a fixed ε). From (67) we deduce that the sequence Y_n is bounded in the space Y_T . Thus we can extract a subsequence that converges weakly in Y_T to $\bar{Y}$ and strongly in X_T by dint of (68). Likewise, thanks to (64) we can assume that v_n converges weakly to $\bar{v}$. So $\bar{Y}$ is then the solution of (57) associated to $\bar{Y}$ and $\bar{v}$. Therefore, in order to show that $K_\varepsilon(\bar{Y}) = \bar{Y}$ it suffices to prove that $\bar{v} = v^{\varepsilon, \bar{Y}}$. From the definition of v_n , we have for all v in $L^2(Q)^m$

$$\begin{aligned} \frac{1}{2} \int_0^T \|v_n\|_{(L^2(0,1))^m}^2 dt + \frac{1}{2\varepsilon} \|Y_n(T)\|_{(L^2(0,1))^n}^2 \\ \leq \frac{1}{2} \int_0^T \|v\|_{(L^2(0,1))^m}^2 dt + \frac{1}{2\varepsilon} \|Y^{\tilde{Y}_n, v}(T)\|_{(L^2(0,1))^n}^2, \end{aligned} \quad (69)$$

where $Y^{\tilde{Y}_n, v}$ is the solution of (57) associated to $\tilde{Y}_n$ and v . Passing to the limit in the inequality (69), one has for all v in $L^2(Q)^m$

$$\begin{aligned} \frac{1}{2} \int_0^T \|\bar{v}\|_{(L^2(0,1))^m}^2 dt + \frac{1}{2\varepsilon} \|\bar{Y}(T)\|_{(L^2(0,1))^n}^2 \\ \leq \frac{1}{2} \int_0^T \|v\|_{(L^2(0,1))^m}^2 dt + \frac{1}{2\varepsilon} \|Y^{\bar{Y}, v}(T)\|_{(L^2(0,1))^n}^2. \end{aligned}$$

This means that $\bar{v}$ minimizes $J_{\varepsilon, \bar{Y}}$. Consequently $K_\varepsilon(\bar{Y}) = \bar{Y}$, Hence the continuity of K_ε . Thus, the following result is then proved.

Theorem 6.1. *Assume the condition (56) is fulfilled. For all Y_0 in $(H_a^1)^n$ the semi-linear parabolic degenerate system (55) is approximatively null controllable. i.e. for all $\varepsilon > 0$ there exists a control $v_\varepsilon \in L^2(Q)^m$ for which the associated solution Y^{v_ε} satisfies*

$$\|Y^{v_\varepsilon}(T)\|_{(L^2(0,1))^n} \leq \varepsilon. \quad (70)$$

Moreover, there exists a positive constant $C > 0$ such that

$$\|v_\varepsilon\|_{L^2(Q)^m}^2 \leq C \|Y_0\|_{(L^2(0,1))^n}^2$$

From this theorem, we deduce following result

Theorem 6.2. *Assume the condition (56) is fulfilled. For all Y_0 in $(H_a^1)^n$ the semi-linear parabolic degenerate system (55) is null controllable. i.e. There exists a control v in $L^2(Q)^m$ for which the associated solution Y^v satisfies*

$$Y^v(T, x) = 0, \quad \forall x \in (0, 1).$$

Moreover, there exists a positive constant $C > 0$ such that

$$\|v\|_{L^2(Q)^m}^2 \leq C \|Y_0\|_{(L^2(0,1))^n}^2.$$

Proof. From Theorem 6.1 the set $\{v_\varepsilon, \varepsilon > 0\}$ is bounded in $L^2(Q)^m$, thus it contains a sequence (v_ε^n) that converges (weakly) in $L^2(Q)^m$ to a limit v_0 that satisfies

$$\|v_0\|_{L^2(Q)^m}^2 \leq C \|Y_0\|_{(L^2(0,1))^n}^2.$$

The sequence $(Y^{v_\varepsilon^n})$ converges strongly to (Y^{v_0}) in X_T . Moreover (Y^{v_0}) is the solution to (55) with $v = v_0$. So according to (70), for all $x \in (0, 1)$ we have

$$Y^{v_0}(T, x) = 0.$$

Thus, the semi-linear parabolic degenerate system (55) with regular initial data is null controllable. $\square$

Now we are able to give the proof of the null controllability of the semi-linear parabolic degenerate system (55) with general initial data. As in [10, 4, 1], we can show also the following well posedness of degenerate parabolic semi-linear systems which is of great utility.

Proposition 6. *For all Y_0 in $(L^2(0, 1))^n$ the semi-linear system*

$$\begin{cases} \partial_t U = \mathbf{D} \mathcal{M} U + F(U) & \text{in } Q, \\ \mathbf{C} U = 0 & \text{on } \Sigma, \\ U(0, x) = Y_0(x) & \text{in } (0, 1), \end{cases} \quad (71)$$

admits a solution U in X_T .

Theorem 6.3. *For all Y_0 in $(L^2(0, 1))^n$ the semi-linear system parabolic degenerate system (55) is null controllable.*

Proof. By Proposition 6 the system (71) in the set $(0, T/2) \times (0, 1)$ with initial data Y_0 in $(L^2(0, 1))^n$ admits a solution $\tilde{Y}$ in $X_{T/2}$. Thus, for $t_0 \in (0, T/2)$ we have $\tilde{Y}(t_0) \in (H_a^1)^n$. Now let us consider

$$\begin{cases} \partial_t \tilde{U} = \mathbf{D}\mathcal{M}\tilde{U} + F(\tilde{U}) + Bv\mathbb{1}_\omega & \text{in } Q_{t_0}, \\ \mathbf{C}\tilde{U} = 0 & \text{on } \Sigma, \\ \tilde{U}(t_0, x) = \tilde{Y}(t_0)(x) & \text{in } (0, 1), \end{cases} \quad (72)$$

where $Q_{t_0} = (t_0, T) \times (0, 1)$. Due to Theorem 6.2, there exists a control $v_1 \in L^2(Q_{t_0})^m$ for which the system (72) admits a solution $\tilde{U}$ that satisfies $\tilde{U}(T, x) = 0$ for all $x \in (0, 1)$. Now let us define (Y, v) as follow :

$$Y = \begin{cases} U & \text{in } [0, t_0], \\ \tilde{U} & \text{in } [t_0, T]. \end{cases}$$

$$v = \begin{cases} 0 & \text{in } [0, t_0], \\ v_1 & \text{in } [t_0, T]. \end{cases}$$

Y is then a solution of the system (55) that satisfies $Y(T, x) = 0$ for all $x \in (0, 1)$ which completes the proof. $\square$

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0.3.2 Controllability of a backward stochastic cascade system of coupled parabolic heat equations by one control force



CONTROLLABILITY OF A BACKWARD STOCHASTIC CASCADE SYSTEM OF COUPLED PARABOLIC HEAT EQUATIONS BY ONE CONTROL FORCE

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ABSTRACT. This paper is devoted to the null controllability of a cascade system of m coupled backward stochastic heat equations governed by a unique distributed control force, where $m \geq 2$. This task is obtained by means of proving a suitable observability estimate for the dual system of the controlled system.

1. Introduction and main result. Let $T > 0$, and $\mathcal{D} \subset \mathbb{R}^N$ be a bounded open set with $\partial\mathcal{D}$ boundary of class $\mathcal{C}^2$. Let $G \subset \mathcal{D}$ be a non empty open subset and $Q = (0, T) \times \mathcal{D}$, $\Sigma = (0, T) \times \partial\mathcal{D}$. Let m be a given fixed integer such that $m \geq 1$, we consider the following backward stochastic system

$$\left\{ \begin{array}{ll} \begin{aligned} d_t y_1 &= \left[-\Delta y_1 + \sum_{i=1}^m a_{1i} y_i + c_1 Y_1 + \mathbb{1}_G v \right] dt + Y_1 dW(t), \\ d_t y_2 &= \left[-\Delta y_2 + \sum_{i=1}^m a_{2i} y_i + c_2 Y_2 \right] dt + Y_2 dW(t), \\ &\vdots \\ d_t y_m &= \left[-\Delta y_m + \sum_{i=1}^m a_{mi} y_i + c_m Y_m \right] dt + Y_m dW(t), \end{aligned} & \text{in } Q \\ \begin{aligned} y_i &= 0 \quad 1 \leq i \leq m \\ y_i(T, \cdot) &= y_{iT}(\cdot) \quad 1 \leq i \leq m \end{aligned} & \begin{array}{l} \text{on } \Sigma, \\ \text{in } \mathcal{D}, \end{array} \end{array} \right. \quad (1)$$

where the couple (y_i, Y_i) is the state variable for the i th equation of the system $1 \leq i \leq m$. $\mathbb{1}_G$ is the characteristic function of the subset G and v is the control acting on G . Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ be a complete filtered probability space, on which a one-dimensional standard Brownian motion $W(t)_{t \geq 0}$ is defined, so that $\mathcal{F} = \{\mathcal{F}_t\}_{t \geq 0}$ is the natural filtration generated by $W(\cdot)$ augmented by all $\mathbb{P}$ -null sets in $\mathcal{F}$.

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Let H be a Banach space, we denote by $L^2_{\mathcal{F}}(0, T; H)$ the Banach space consisting of all H -valued $\{\mathcal{F}_t\}_{t \geq 0}$ -adapted processes $X(\cdot)$ such that $\mathbb{E}(|X(\cdot)|^2_{L^2(0, T; H)}) < \infty$, with the canonical norm; by $L^\infty_{\mathcal{F}}(0, T; H)$ the Banach space consisting of all H -valued $\{\mathcal{F}_t\}_{t \geq 0}$ -adapted bounded processes; by $L^2_{\mathcal{F}}(\Omega; C([0, T]; H))$ the Banach space consisting of all H -valued $\{\mathcal{F}_t\}_{t \geq 0}$ -adapted processes $X(\cdot)$ such that $\mathbb{E}(|X(\cdot)|^2_{C(0, T; H)}) < \infty$, with the canonical norm.

Let us denote by $y = (y_1, \dots, y_m)^*$ the first variable state, $Y = (Y_1, \dots, Y_m)^*$ the second variable state and $y_T = (y_{1T}, \dots, y_{mT})^*$ the initial datum. The matrices

$$A = (a_{i,j})_{1 \leq i, j \leq m}, \quad C = \text{diag}(c_1, \dots, c_m) \text{ and } B = (1, 0, \dots, 0)^*.$$

Thus the system (1) can be written as

$$\begin{cases} d_t y = \left(-\tilde{\Delta} y + Ay + CY + B \mathbb{1}_G v \right) dt + Y dW(t), & \text{in } Q \\ y = 0 & \text{on } \Sigma, \\ y(T) = y_T(\cdot) & \text{in } \mathcal{D}, \end{cases} \quad (1.1)$$

where $\tilde{\Delta}$ is the matrix operator given by $\tilde{\Delta} = \text{diag}(\Delta, \dots, \Delta)$.

Null controllability of coupled systems has been widely studied in the last half-century, and a great number of results have been established. In deterministic case M. Gonzalez-Burgos, L. de Teresa [9] provided a null controllability result for a cascade parabolic system, F. Ammar-Khodja et al. [4] obtained several results characterizing the null controllability of fully coupled systems with m -control forces by a generalized Kalman rank condition. For degenerate systems we can also cite [1, 2, 3, 8] and references therein.

In stochastic case, the controllability of forward stochastic differential equation (FSDE) which is equivalent with the observability of backward stochastic differential equation (BSDE) is still an open problem [5]. However, the controllability results for one backward stochastic equation were initiated by Barbu, Răscanu and Tessitore [6], later others results were obtained by Tang and Zhang [17]. Whereas, there are only few results relating to coupled systems Hongheng Li and Qi Lu [12] established the null controllability for a systems of two backward stochastic heat equations with one control force. To our knowledge, the authors in [14] transform a control problem sensitizing to the controllability of a coupled system of a forward equation and a backward equation.

In both cases, deterministic and stochastic, the key ingredient for proving the null controllability is to obtain the observability inequality for the corresponding adjoint system. The observability inequality is basically obtained by means of estimates which are called Carleman estimates.

In this paper, we are interested in the null controllability of a backward system of m coupled parabolic equations ($m \geq 2$) under the action of one control force knowing that all coefficients of the sub-diagonal in the coupling matrix A are non-zero. Recall that the null controllability of system (1) is formulated as follows:

Definition 1.1. System (1) is said to be null controllable at time $T > 0$ if for any given $y_T = (y_{1T}, \dots, y_{mT}) \in L^2(\Omega, \mathcal{F}_T, \mathbb{P}; (L^2(D))^m)$, one can find a control $v \in L^2_{\mathcal{F}}(0, T; L^2(G))$ such that the solution $((y_1, Y_1), \dots, (y_m, Y_m))$ of system (1) satisfies $y_1(0) = \dots = y_m(0) = 0$ in D , $\mathbb{P}$ -a.s.

In the sequel, we assume that the coupling first state variable matrix A have the following structure

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2m} \\ 0 & a_{32} & a_{33} & \cdots & a_{3m} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{m,m-1} & a_{mm} \end{pmatrix} \quad (2)$$

with $a_{ij} \in L^\infty(Q)$, $B_{ij} \in L^\infty(Q)^N$ ($1 \leq i \leq j \leq m$) and $a_{i,i-1} \in L^\infty(Q)$ ($2 \leq i \leq m$), for an open set $G_0 \subset G$, satisfy

$$a_{i,i-1} \geq a_0 > 0 \text{ or } -a_{i,i-1} \geq a_0 > 0 \text{ in } G_0 \times (0, T), \quad \forall i : 2 \leq i \leq m, \quad (3)$$

and the second coupling matrix C satisfies

$$c_i \in L^\infty(Q) \quad (4)$$

At present, under the previous conditions, the main result about the controllability of the system is the following

Theorem 1.2. *Assume that the coupling matrices A and C satisfy the assumptions (2)-(4), given $y_T = (y_{1T}, \dots, y_{mT}) \in L^2(\Omega, \mathcal{F}_T, \mathbb{P}; (L^2(D))^m)$. There exist a control $v \in L^2_{\mathcal{F}}(0, T; L^2(G))$ such that the solution $((y_1, Y_1), \dots, (y_m, Y_m))$ of system (1) satisfies $y_1(0) = \dots = y_m(0) = 0$ in D , $\mathbb{P}$ -a.s.*

The rest of this paper is organized as follows. Section 2 is devoted to well-posedness of both backward and forward stochastic parabolic equations. In section 3, we give Carleman estimates for the forward stochastic degenerate parabolic equation with multiplicative noise. And in the last section, we establish the observability inequality for the adjoint system (17) corresponding to the initial controlled system (1) (see Proposition 4.3), we deduce the unique continuation property the adjoint system (17) (see Corollary 4.4) and finally we deduce the proof of the main Theorem 1.2.

All along the article, we use generic constants for the estimates, whose values may change from line to line.

2. Systems of coupled forward/backward equations. In the light of the results established for a single equation (see [6]) we therefore infer the following results on systems of coupled forward/backward stochastic equations

$$\begin{cases} d_t y(t, x) = \Delta y(t, x) + A(t, x)y(t, x)dt + F(t, x)dt + b(t, x)y(t, x)dW(t) & \text{in } Q, \\ y(t, x) = 0 & \text{on } \Sigma, \\ y(0, x) = y_0(x) & \text{in } \mathcal{D}, \end{cases} \quad (5)$$

where

- (i) $F \in L^2_{\mathcal{F}}(\Omega; L^2([0, T]; L^2(\mathcal{D})^m))$,
- (ii) $A \in L^{\infty}_{\mathcal{F}}(\Omega; L^{\infty}([0, T]; L^{\infty}(\mathcal{D})^m))$, $b \in L^{\infty}_{\mathcal{F}}(\Omega; L^{\infty}([0, T]; L^{\infty}(\mathcal{D})))$,
- (iii) $y_0 \in (L^2(\mathcal{D}))^m$.

Under the assumptions (6) equation (5) has a unique solution y belonging to $(L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D})^m)))$. Moreover, if $y_0 \in (H^1_0(\mathcal{D}))^m$ and $b \in L^{\infty}_{\mathcal{F}}(\Omega; L^{\infty}([0, T]; W^{1,\infty}(\mathcal{D})))$ then $y(t)$ is in $(H^1_0(\mathcal{D}) \cap H^2(\mathcal{D}))^m$, $\mathbb{P}$ -almost surely for every $t \in]0, T]$ and Δy belongs to $(L^2(\Omega; L^2(0, T; L^2(\mathcal{D})^m)))$.

We also consider the backward equation

$$\begin{cases} dp(t, x) = [-\Delta p(t, x) - Ap(t, x) + b(t, x)k(t, x)dt + G(t, x)]dt + k(t, x)dW(t) & \text{in } Q, \\ p(t, x) = 0 & \text{on } \Sigma, \\ p(T, x) = p_T(x) & \text{in } \mathcal{D}, \end{cases} \quad (7)$$

where

$$\begin{aligned} (i) \quad & G \in (L^2_{\mathcal{F}}(\Omega; L^2(0, T; L^2(\mathcal{D}))))^m, \\ (ii) \quad & p_T \in (L^2(\Omega, \mathcal{F}_T, \mathbb{P}; L^2(\mathcal{D})))^m. \end{aligned} \quad (8)$$

Under the previous conditions, the backward stochastic differential system (7) has a unique solution (p, k) with $p \in (L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D})^m)) \cap L^2_{\mathcal{F}}(\Omega; L^2([0, T]; H_0^1(\mathcal{D})^m)))$ and $k \in (L^2_{\mathcal{F}}(\Omega; L^2([0, T]; L^2(\mathcal{D})^m)))$. Moreover, if $p_T \in L^2(\Omega, \mathcal{F}_T, \mathbb{P}; H_0^1(\mathcal{D})^m)$ and $b \in L^{\infty}_{\mathcal{F}}(\Omega, L^{\infty}([0, T], W^{1,\infty}(\mathcal{D})))$, then we have :

$$\begin{aligned} p &\in L^2_{\mathcal{F}}(\Omega; L^2([0, T]; H^2(\mathcal{D})^m)) \cap L^2_{\mathcal{F}}(\Omega; C([0, T]; H_0^1(\mathcal{D})^m)) \\ \text{and } k &\in L^2_{\mathcal{F}}(\Omega; L^2([0, T]; H_0^1(\mathcal{D})^m)). \end{aligned}$$

3. Carleman estimate for a forward stochastic parabolic equation. First of all, let us introduce the weighted functions necessary for the Carleman estimation, for this we recall the following result (see [17, Lemma 5.1])

Lemma 3.1. *Let $G_0 \Subset G$ be an arbitrary fixed subdomain of $\mathcal{D}$. Then there exists a function $\psi \in C^4(\overline{\mathcal{D}})$ such that*

$$\psi(x) > 0 \quad \forall x \in \mathcal{D}, \quad \psi|_{\partial\mathcal{D}} = 0, \quad |\nabla\psi(x)| > 0 \quad \forall x \in \overline{\mathcal{D}} \setminus G_0.$$

Put

$$\alpha(t, x) = \frac{e^{\lambda\psi(x)} - e^{2\lambda|\psi|_{L^{\infty}(\mathcal{D})}}}{t^2(T-t)^2}, \quad \varphi(t, x) = \frac{e^{\lambda\psi(x)}}{t^2(T-t)^2} \quad \text{and} \quad \theta(t) = \frac{1}{t^2(T-t)^2} \quad (9)$$

Therefore, we have the following estimate of forward stochastic heat equations (see [13, Theorem 1.1]).

Lemma 3.2. *For any $T > 0$, there exists a constant $\lambda_0 > 0$ such that for all $\lambda \geq \lambda_0$, one can find two constants $C = C(\lambda) > 0$ and $s_0 = s_0(\lambda) > 0$ so that for all $p \in L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D}))) \cap L^2_{\mathcal{F}}(0, T; H_0^1(\mathcal{D}))$, $f_i \in L^2_{\mathcal{F}}(0, T; L^2(\mathcal{D}))$ ($i = 1, 2$) satisfying*

$$dp - \Delta p \, dt = f_1 \, dt + f_2 \, dW(t) \quad (10)$$

it holds that

$$\begin{aligned} \mathbb{E} \int_Q e^{2s\alpha} (s^3 \theta^3 p^2 + s\theta |\nabla p|^2) \, dxdt &\leq C \left(\mathbb{E} \int_0^T \int_G s^3 \theta^3 e^{2s\alpha} p^2 \, dxdt \right. \\ &\quad \left. + \mathbb{E} \int_Q e^{2s\alpha} f_1^2 \, dxdt + \mathbb{E} \int_Q s^2 \theta^2 f_2^2 e^{2s\alpha} \, dxdt \right) \end{aligned} \quad (11)$$

for any $s \geq s_0$.

Now, consider the following forward stochastic equation with multiplicative nose

$$dy = \Delta y \, dt + f \, dt + by \, dW(t) \quad (12)$$

Applying Lemma 3.2 to (12), the last term in the right hand-side of (11) can be absorbed by the first term in the left hand side for s large enough. Thus we get the following result

Corollary 3.3. *There exists two constants $C' > 0$ and $s_0 > 0$ so that for all $f \in L^2_{\mathcal{F}}(0, T; L^2(\mathcal{D}))$ and $y \in L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D}))) \cap L^2_{\mathcal{F}}(0, T; H^1_0(\mathcal{D}))$, satisfying (12), it holds that*

$$\begin{aligned} & \mathbb{E} \int_Q ((s\theta)^1 |\nabla y|^2 + (s\theta)^3 y^2) e^{2s\alpha} dxdt \\ & \leq C' \left\{ \mathbb{E} \int_Q f^2 e^{2s\alpha} dxdt + \mathbb{E} \int_0^T \int_G (s\theta)^3 y^2 e^{2s\alpha} dxdt \right\} \end{aligned} \quad (13)$$

for all $s \geq s_0$, where the constant $C' > 0$ is independent of s .

Now, we can generalize the estimate (13) using a similar technique as in [11] to get the following result

Lemma 3.4. *Let $d \in \mathbb{R}$, there exists two constants $C' > 0$ and $s_0 > 0$ so that for all $f \in L^2_{\mathcal{F}}(0, T; L^2(\mathcal{D}))$ and $y \in L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D}))) \cap L^2_{\mathcal{F}}(0, T; H^1_0(\mathcal{D}))$, satisfying (12), it holds that*

$$\begin{aligned} & \mathbb{E} \int_Q ((s\theta)^{1+d} |\nabla y|^2 + (s\theta)^{3+d} y^2) e^{2s\alpha} dxdt \\ & \leq C \left\{ \mathbb{E} \int_Q \theta^d f^2 e^{2s\alpha} dxdt + \mathbb{E} \int_0^T \int_G (s\theta)^{3+d} y^2 e^{2s\alpha} dxdt \right\} \end{aligned} \quad (14)$$

for all $s \geq s_0$, where the constant $C > 0$ is independent of s .

Proof. In the case of $d = 0$, inequality (14) is shown in Corollary 3.3. For $d \neq 0$, set $w = y(s\theta)^{\frac{d}{2}}$. Thus we have

$$dw = [\Delta w + (s\theta)^{\frac{d}{2}} f + \frac{d}{2} \theta' \theta^{-1} w] dt + b w dW(t) \quad (15)$$

Now, applying Corollary 3.3 to the equation (15) we get

$$\begin{aligned} & \mathbb{E} \int_Q ((s\theta)^1 |\nabla w|^2 + (s\theta)^3 w^2) e^{2s\alpha} dxdt \\ & \leq C \left\{ \mathbb{E} \int_Q ((s\theta)^{\frac{d}{2}} f + \frac{d}{2} \theta' \theta^{-1} w)^2 e^{2s\alpha} dxdt \right. \\ & \quad \left. + \mathbb{E} \int_0^T \int_G (s\theta)^3 w^2 e^{2s\alpha} dxdt \right\}. \end{aligned}$$

Knowing that $|\theta' \theta^{-1}| \leq 2T\theta^{\frac{1}{2}}$, the corresponding term can then be absorbed by the second term in the left hand-side for s large enough. Therefore we get

$$\begin{aligned} & \mathbb{E} \int_Q ((s\theta)^1 |\nabla w|^2 + (s\theta)^3 w^2) e^{2s\alpha} dxdt \leq C \left\{ \mathbb{E} \int_Q (s\theta)^d f^2 e^{2s\alpha} dxdt \right. \\ & \quad \left. + \mathbb{E} \int_0^T \int_G (s\theta)^3 w^2 e^{2s\alpha} dxdt \right\}, \end{aligned} \quad (16)$$

for all $s \geq s_0$.

Consequently, the change of $w = y(s\theta)^{\frac{d}{2}}$ in the estimate (16) yields (14). $\square$

4. **Proof of the main result.** The question of controllability of system (1) leads us to consider its corresponding conjoint system (17) defined as follows

$$\begin{cases} dp_1 = [\Delta p_1 - a_{11}p_1 - a_{21}p_2] dt - c_1 p_1 dW(t), \\ \vdots \\ dp_i = \left[\Delta p_i - \sum_{j=1}^i a_{ji}p_j - a_{i+1,i}p_{i+1} \right] dt - c_i p_i dW(t), & \text{in } Q, \\ \vdots \\ dp_m = \left[\Delta y_m - \sum_{j=1}^m a_{jm}p_j \right] dt - c_m p_m dW(t), \\ p_i = 0 \quad 1 \leq i \leq m & \text{on } \Sigma, \\ p_i(0, \cdot) = p_{i0}(\cdot) \quad 1 \leq i \leq m & \text{in } \mathcal{D}, \end{cases} \quad (17)$$

in matrix writing and using the same notations as system (1)

$$\begin{cases} d_t p = (\tilde{\Delta} p - A^* p) dt - C p dW(t), & \text{in } Q \\ p = 0 & \text{on } \Sigma, \\ p(0) = y_0(\cdot) & \text{in } \mathcal{D}, \end{cases} \quad (4.1)$$

where $p = (p_1, \dots, p_m)^*$.

Let d in $\mathbb{R}$ a real number, $z \in L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D}))) \cap L^2_{\mathcal{F}}(0, T; H^1_0(\mathcal{D}))$ and $B \subset \mathcal{D}$ in order to lighten the notations, we set

$$\mathcal{J}(d, z) = \mathbb{E} \int_Q ((s\theta)^{d-2} |\nabla z|^2 + (s\theta)^d z^2) e^{2s\alpha} dx dt$$

and

$$\mathcal{L}_B(d, z) = \mathbb{E} \int_{B \times (0, T)} (s\theta)^d z^2 e^{2s\alpha} dx dt$$

Arguing like in the deterministic case [9], the following result is strongly requested in the proof of the main theorem.

Lemma 4.1. *Under assumptions of Theorem 1.2 and given $l \in \mathbb{N}, \varepsilon > 0, k \in \{2, \dots, m\}$ and two open sets $\mathcal{O}_0$ and $\mathcal{O}_1$ such that $G_1 \subset \mathcal{O}_1 \subset \subset \mathcal{O}_0 \subset G_0$, there exist a positive constant C_k and $l_{kj} \in \mathbb{N}, 1 \leq j \leq k-1$, such that if $p = (p_1, \dots, p_m)^*$ is the solution to (17) associated to $p_0 = (p_{10}, \dots, p_{m0})^* \in L^2(Q)^m$ and $s \geq s_0$, one has*

$$\begin{aligned} \mathcal{L}_{\mathcal{O}_1}(l, p_k) &\leq \varepsilon [\mathcal{J}(3(m+1-k), p_k) + \mathcal{J}(3(m-k), p_{k+1})] \\ &\quad + C_k \left(1 + \frac{1}{\varepsilon}\right) \sum_{j=1}^{k-1} \mathcal{L}_{\mathcal{O}_0}(l_{k,j}, p_j). \end{aligned} \quad (18)$$

(In this inequality we have taken $p_{m+1} \equiv 0$).

Proof. Given $G_1 \subset \mathcal{O}_1 \Subset \mathcal{O}_0 \subset G_0$, we consider $\xi \in \mathcal{C}^\infty(\mathbb{R}^N)$ such that $0 \leq \xi \leq 1$ in $\mathbb{R}^N$, $\xi \equiv 1$ in $\mathcal{O}_1$, $\text{Supp}(\xi) \subset \mathcal{O}_0$ and

$$\frac{\Delta \xi}{\xi^{1/2}} \in L^\infty(\mathcal{D}) \quad \text{and} \quad \frac{\nabla \xi}{\xi^{1/2}} \in L^\infty(\mathcal{D})^N \quad (19)$$

Let us consider $k \in \{2, \dots, m\}$, knowing that the coefficient $a_{k,k-1}$ satisfies hypothesis (3) we assume without loss of generality that $a_{k,k-1} \geq a_0 > 0$ in $G_0 \times (0, T)$. Let $\ell \in \mathbb{N}$ and consider $u = (s\theta)^\ell e^{2s\alpha}$. Differentiating $d_s(p_k \cdot p_{k-1})$, multiplying by $u\xi$, integrating on Q and computing the mean value, we obtain

$$\begin{aligned} a_0 \mathcal{L}_{\mathcal{O}_1}(\ell, p_k) &\leq \mathbb{E} \int_Q a_{k,k-1} u \xi p_k^2 dx dt \\ &= \mathbb{E} \int_Q \left(u \xi p_k \Delta p_{k-1} - \sum_{j=1}^{k-1} a_{j,k-1} u \xi p_k p_j \right) dx dt \\ &\quad + \mathbb{E} \int_Q \left(u \xi p_{k-1} \Delta p_k - \sum_{j=1}^k a_{j,k} u \xi p_{k-1} p_j - a_{k+1,k} u \xi p_{k+1} p_{k-1} \right) dx dt \\ &\quad + \mathbb{E} \int_Q u \xi c_{k-1} c_k p_{k-1} p_k dx dt. \end{aligned}$$

Thanks to green formula and taking the absolute value, we get

$$\begin{aligned} a_0 \mathcal{L}_{\mathcal{O}_1}(\ell, p_k) &\leq \mathbb{E} \int_Q \nabla(u\xi) \nabla p_{k-1} p_k dx dt + \mathbb{E} \int_Q \nabla(u\xi) \nabla p_k p_{k-1} dx dt \\ &\quad + \mathbb{E} \int_Q u \xi \nabla p_k \nabla p_{k-1} dx dt + \sum_{j=1}^{k-1} \mathbb{E} \int_Q a_{j,k-1} u \xi p_k p_j dx dt \\ &\quad + \sum_{j=1}^k \mathbb{E} \int_Q a_{j,k} u \xi p_{k-1} p_j dx dt + \mathbb{E} \int_Q a_{k+1,k} u \xi p_{k+1} p_{k-1} dx dt \\ &\quad + \mathbb{E} \int_Q u \xi c_{k-1} c_k p_{k-1} p_k dx dt. \\ &= \sum_{i=1}^7 I_i. \end{aligned}$$

Since

$$|\nabla(u)| \leq C(s\theta)^{l+1} e^{2s\alpha}, \quad (20)$$

then we have

$$\begin{aligned} I_1 &= \mathbb{E} \int_Q \nabla(u\xi) \nabla p_{k-1} p_k dx dt \\ &\leq C \mathbb{E} \int_Q \left(\sqrt{\frac{2\varepsilon}{C}} (s\theta)^{\frac{3(m+1-k)}{2}} e^{s\alpha} p_k \right) \left(\sqrt{\frac{C}{2\varepsilon}} (s\theta)^{-\frac{3(m+1-k)}{2}} e^{-s\alpha} \nabla(u\xi) \nabla p_{k-1} \right) dx dt \\ &\leq \varepsilon \mathcal{J}(3(m+1-k), p_k) + \frac{C^2}{4\varepsilon} \mathbb{E} \int_Q (s\theta)^{2\ell+2-3(m-k+1)} e^{2s\alpha} \xi |\nabla p_{k-1}|^2 dx dt. \quad (21) \end{aligned}$$

Thanks to Young inequality and using (19) and (20) we get

$$I_2 = \mathbb{E} \int_Q \nabla(u\xi) \nabla p_k p_{k-1} dx dt$$

$$\begin{aligned}
&\leq C \mathbb{E} \int_Q \left(\sqrt{\frac{2\varepsilon}{C}} (s\theta)^{\frac{3(m+1-k)-2}{2}-\ell-1} e^{-s\alpha} \xi^{-1/2} \nabla(u\xi) \nabla p_k \right) \\
&\quad \times \left(\sqrt{\frac{C}{2\varepsilon}} (s\theta)^{\ell+1-\frac{3(m+1-k)-2}{2}} e^{s\alpha} \xi^{1/2} p_{k-1} \right) dx dt \\
&\leq \varepsilon \mathcal{J}(3(m+1-k), p_k) + \frac{C^2}{4\varepsilon} \mathcal{L}_{\mathcal{O}_0}(2\ell-3(m+1-k), p_{k-1}). \quad (22)
\end{aligned}$$

Using similar computations, we get

$$\begin{aligned}
I_3 &= \mathbb{E} \int_Q u \xi \nabla p_k \nabla p_{k-1} dx dt \\
&\leq \varepsilon \mathbb{E} \int_Q (s\theta)^{3(m-k+1)-2} e^{2s\alpha} \xi |\nabla p_k|^2 dx dt \\
&\quad + \frac{C^2}{4\varepsilon} \mathbb{E} \int_Q (s\theta)^{2\ell+2-3(m-k+1)} e^{2s\alpha} \xi |\nabla p_{k-1}|^2 dx dt \\
&\leq \varepsilon \mathcal{J}(3(m+1-k), p_k) + \frac{C^2}{4\varepsilon} \mathbb{E} \int_Q (s\theta)^{2\ell+2-3(m-k+1)} e^{2s\alpha} \xi |\nabla p_{k-1}|^2 dx dt. \quad (23)
\end{aligned}$$

$$\begin{aligned}
I_4 &= \sum_{j=1}^{k-1} \mathbb{E} \int_Q a_{j,k-1} u \xi p_k p_j dx dt \leq \sum_{j=1}^{k-1} C \mathbb{E} \int_Q (s\theta)^\ell e^{2s\alpha} \xi p_k p_j dx dt \\
&\leq \sum_{j=1}^{k-1} C \mathbb{E} \int_Q \left(\sqrt{\frac{2\varepsilon}{(k-1)C}} (s\theta)^{\frac{3(m+1-k)}{2}} e^{s\alpha} \sqrt{\xi} p_k \right) \\
&\quad \times \left(\sqrt{\frac{(k-1)C}{2\varepsilon}} (s\theta)^{\ell-\frac{3(m+1-k)}{2}} e^{s\alpha} \sqrt{\xi} p_j \right) dx dt \\
&\leq \varepsilon \mathbb{E} \int_Q (s\theta)^{3(m+1-k)} e^{2s\alpha} \xi p_k^2 dx dt \\
&\quad + \sum_{j=1}^{k-1} \frac{(k-1)C^2}{4\varepsilon} \mathbb{E} \int_Q (s\theta)^{2\ell-3(m+1-k)} e^{2s\alpha} \xi p_j^2 dx dt \\
&\leq \varepsilon \mathcal{J}(3(m+1-k), p_k) + \sum_{j=1}^{k-1} \frac{(k-1)C^2}{4\varepsilon} \mathcal{L}_{\mathcal{O}_0}(2\ell-3(m+1-k), p_j). \quad (24)
\end{aligned}$$

Again, using Young inequality

$$I_5 = \sum_{j=1}^{k-2} \mathbb{E} \int_Q a_{j,k} u \xi p_{k-1} p_j dx dt + \mathbb{E} \int_Q a_{j,k} u \xi p_{k-1}^2 dx dt + \mathbb{E} \int_Q a_{j,k} u \xi p_{k-1} p_k dx dt. \quad (25)$$

Thanks to Holder inequality, the first term can be bounded as follows :

$$\sum_{j=1}^{k-2} \mathbb{E} \int_Q a_{j,k} u \xi p_{k-1} p_j dx dt \leq \sum_{j=1}^{k-2} C \mathbb{E} \int_Q (s\theta)^\ell e^{2s\alpha} \xi p_{k-1} p_j dx dt \leq C \sum_{j=1}^{k-1} \mathcal{L}_{\mathcal{O}_0}(\ell, p_j).$$

The second term in (25) can be bounded by $\mathcal{L}_{\mathcal{O}_1}(\ell, p_{k-1})$ and the last one can be bounded like I_7 . Thus we infer

$$I_5 \leq \varepsilon \mathcal{I}(3(m+1-k), p_k) + C \sum_{j=1}^{k-1} \mathcal{L}_{\mathcal{O}_0}(\ell_j, p_j) \quad (26)$$

Likewise for the next term we have

$$\begin{aligned} I_6 &\leq C \mathbb{E} \int_Q (s\theta)^\ell e^{2s\alpha} \xi p_{k-1} p_{k+1} dx dt \\ &\leq C \mathbb{E} \int_Q \left(\sqrt{\frac{2\varepsilon}{C}} (s\theta)^{\frac{3(m-k)}{2}} e^{s\alpha} \sqrt{\xi} p_{k+1} \right) \left(\sqrt{\frac{C}{2\varepsilon}} (s\theta)^{\ell - \frac{3(m-k)}{2}} e^{s\alpha} \sqrt{\xi} p_{k-1} \right) dx dt \\ &\leq \varepsilon \mathcal{I}(3(m-k), p_{k+1}) + \frac{C^2}{4\varepsilon} \mathcal{L}_{\mathcal{O}_0}(2\ell - 3(m-k), p_{k-1}) \end{aligned} \quad (27)$$

Arguing in a similar way like I_6 we get

$$I_7 \leq \varepsilon \mathcal{I}(3(m+1-k), p_k) + \frac{C}{4\varepsilon} \mathcal{L}_{\mathcal{O}_0}(2\ell - 3(m+1-k), p_{k-1}). \quad (28)$$

Altogether from (21)-(28), we obtain

$$\begin{aligned} \mathcal{L}_{\mathcal{O}_1}(\ell, p_k) &\leq \varepsilon [\mathcal{I}(3(m+1-k), p_k) + \mathcal{I}(3(m-k), p_{k+1})] \\ &\quad + C_k \left(1 + \frac{1}{\varepsilon} \right) \sum_{j=1}^{k-1} \mathcal{L}_{\mathcal{O}_0}(\ell_{kj}, p_j) \\ &\quad + \frac{C^2}{4\varepsilon} \mathbb{E} \int_Q (s\theta)^{2\ell+2-3(m-k+1)} e^{2s\alpha} \xi |\nabla p_{k-1}|^2 dx dt. \end{aligned} \quad (29)$$

The next task is to absorb the term $\mathbb{E} \int_Q (s\theta)^R e^{2s\alpha} \xi |\nabla p_{k-1}|^2 dx dt$. Let $\bar{u} = (s\theta)^R e^{2s\alpha}$, multiplying the equation satisfied by p_{k-1} by $\bar{u} \xi p_{k-1}$, integrating on Q and computing the expectation we get

$$\begin{aligned} \mathbb{E} \int_Q \bar{u} \xi |\nabla p_{k-1}|^2 dx dt &\leq \mathbb{E} \int_Q \nabla(\bar{u} \xi) \nabla p_{k-1} p_{k-1} dx dt + \sum_{j=1}^{k-2} \mathbb{E} \int_Q a_{j,k-1} \bar{u} \xi p_{k-1} p_j dx dt \\ &\quad + \mathbb{E} \int_Q a_{k-1,k-1} \bar{u} \xi p_{k-1}^2 dx dt + \mathbb{E} \int_Q a_{k,k-1} \bar{u} \xi p_{k-1} p_k dx dt \\ &= K_1 + K_2 + K_3 + K_4. \end{aligned} \quad (30)$$

The first term

$$K_1 \leq \frac{1}{2} \mathbb{E} \int_Q \nabla(\bar{u} \xi) \nabla p_{k-1}^2 dx dt \leq C \mathbb{E} \int_Q \Delta(\bar{u} \xi) p_{k-1}^2 dx dt \leq C \mathcal{L}_{\mathcal{O}_0}(R+1, p_{k-1}) \quad (31)$$

The second term

$$\begin{aligned} K_2 &\leq C \sum_{j=1}^{k-2} \mathbb{E} \int_Q (s\theta)^R \xi e^{2s\alpha} p_{k-1} p_j dx dt \\ &\leq C \sum_{j=1}^{k-2} \mathbb{E} \int_Q \left((s\theta)^{\frac{R}{2}} \sqrt{\xi} e^{s\alpha} p_{k-1} \right) \left((s\theta)^{\frac{R}{2}} \sqrt{\xi} e^{s\alpha} p_j \right) dx dt \end{aligned}$$

$$\begin{aligned}
&\leq C \mathbb{E} \int_Q (s\theta)^R \xi e^{2s\alpha} p_{k-1}^2 dx dt + C \sum_{j=1}^{k-2} \mathbb{E} \int_Q (s\theta)^R \xi e^{2s\alpha} p_j^2 dx dt \\
&\leq C \mathcal{L}_{\mathcal{O}_0}(R, p_{k-1}) + \sum_{j=1}^{k-2} C \mathcal{L}_{\mathcal{O}_0}(R, p_j).
\end{aligned} \tag{32}$$

K_3 can be bounded by $C \mathcal{L}_{\mathcal{O}_0}(R, p_{k-1})$. Arguing like the term I_6 we get

$$K_4 \leq \varepsilon \mathcal{J}(3(m+1-k), p_k) + \frac{C}{4\varepsilon} \mathcal{L}_{\mathcal{O}_0}(2R - 3(m+1-k), p_{k-1}). \tag{33}$$

From (30)-(33) we deduce the estimate (18). $\square$

Therefore, we infer the following global Carleman estimate for the adjoint system (17).

Theorem 4.2. *Under assumptions of Theorem 1.2, there exist two positive constants C_0 and s_0 and $l \geq 3$ such that, for every $p_0 = (p_{10}, \dots, p_{m0})^* \in L^2(Q)^m$, the solution $p = (p_1, \dots, p_m)^*$ to (17) satisfies*

$$\sum_{i=1}^m \mathcal{J}(3(m+1-i), p_i) \leq \widetilde{C}_0 \mathbb{E} \left(s^\ell \int_{G_0 \times (0, T)} e^{2s\alpha} \theta^\ell |p_1|^2 dx dt \right), \tag{34}$$

for all $s \geq s_0$.

Proof. Given $G_0 \subset G$, we choose $G_1 \Subset G_0$. Let $\psi \in C^4(\overline{\mathcal{D}})$ be the function provided by Lemma 3.1 and associated to $\mathcal{D}$ and G_1 and let α its associated function given by (9).

Let $p = (p_1, \dots, p_m)^*$ be the solution to (17) associated to $p_0 = (p_{01}, \dots, p_{0m})^* \in (L^2(\mathcal{D}))^m$. We apply Lemme 3.4 to each p_i ($1 \leq i \leq m$) with $d = 3(m+1-i)$ to get

$$\mathcal{J}(3(m+1-i), p_i) \leq C \left(\mathbb{E} \int_Q \theta^{3(m-i)} \left(\sum_{j=1}^{i+1} a_{j,i} p_j \right)^2 e^{2s\alpha} dx dt + \mathcal{L}_{G_1}(3(m+1-i), p_i) \right). \tag{35}$$

For all $j \in \{1, \dots, i+1\}$ we have

$$\mathbb{E} \int_Q \theta^{3(m-i)} a_{j,i}^2 p_j^2 e^{2s\alpha} dx dt \leq \mathcal{J}(3(m+1-j), p_j)$$

For s large enough. Thus

$$\sum_{i=1}^m \mathcal{J}(3(m+1-i), p_i) \leq C_1 \left(\sum_{i=1}^m \mathcal{L}_{G_1}(3(m+1-i), p_i) \right). \tag{36}$$

Let us consider a sequence of open sets $\tilde{\mathcal{O}}_i \subset G_0$ with $2 \leq i \leq m$ such that

$$G_1 \Subset \tilde{\mathcal{O}}_m \Subset \tilde{\mathcal{O}}_{m-1} \Subset \dots \Subset \tilde{\mathcal{O}}_3 \Subset \tilde{\mathcal{O}}_2 \subset G_0.$$

We apply the formula (18) with $\mathcal{O}_1 = G_1$, $\mathcal{O}_0 = \tilde{\mathcal{O}}_m$, $k = m$, $\ell = 3$ and $\varepsilon = \frac{1}{2C_1}$, where C_1 is the constant in (36)

$$\sum_{i=1}^m \mathcal{J}(3(m+1-i), p_i) \leq \widetilde{C}_m \left(\sum_{i=1}^{m-1} \mathcal{L}_{\tilde{\mathcal{O}}_m}(l_{m,i}, p_i) \right) \tag{37}$$

for all $s \geq s_0$ with $\widetilde{C}_m$ a new positive constant. In the formula (37) we have eliminated in the right-hand side the term that depends on p_m . Once again, we

apply the formula (18) with $\mathcal{O}_1 = \tilde{\mathcal{O}}_m$, $\mathcal{O}_0 = \tilde{\mathcal{O}}_{m-1}$, $k = m - 1$, $\ell = l_{m,m-1}$ and $\varepsilon = \frac{1}{2\tilde{C}_m}$ and eliminate in (37) the local term $\mathcal{L}_{\tilde{\mathcal{O}}_m}(l_{m,m-1}, p_{m-1})$. By finite iteration of this argument we obtain the estimate (34). $\square$

The following result is of great interest in the proof of the main theorem as well as in the proof of the unique continuation of the adjoint problem which is a cascade system of coupled forward stochastic equations (17)

Proposition 4.3. *Under assumptions of Theorem 1.2, there exist two positive constants C and s_0 such that, for every $p_0 = (p_{10}, \dots, p_{m0})^* \in L^2(Q)^m$, the solution $p = (p_1, \dots, p_m)^*$ to (17) satisfies*

$$\mathbb{E} \left(\sum_{i=1}^m \int_{\mathcal{D}} p_i^2(T, x) dx \right) \leq C \mathbb{E} \left(s^\ell \int_{G_0 \times (0, T)} e^{2s\alpha\theta^\ell} |p_1|^2 dx dt \right), \quad (38)$$

for all $s \geq s_0$.

Proof. For $i \in \{1, \dots, m\}$. Differentiating $d_\tau \|p_i\|_{L^2(0,1)}^2$, integrating on $[s, t]$ (with $0 \leq s \leq t \leq T$) and taking the mean value, we have

$$\mathbb{E} \int_{\mathcal{D}} p_i^2(t, x) dx \leq \mathbb{E} \int_{\mathcal{D}} p_i^2(s, x) dx + C \mathbb{E} \int_s^t \int_{\mathcal{D}} p_i^2(\ell, x) dx d\ell.$$

Tanks to Gronwall's lemma, we get

$$\mathbb{E} \int_{\mathcal{D}} p_i^2(t, x) dx \leq e^{C(t-s)} \mathbb{E} \int_{\mathcal{D}} p_i^2(s, x) dx \leq e^{CT} \mathbb{E} \int_{\mathcal{D}} p_i^2(s, x) dx.$$

Knowing that $\theta^3(t) e^{s\alpha(t,x)} \geq (\frac{2}{T})^{12} \exp(-se^{2\lambda|\psi|_\infty} \theta)$ for all $(t, x) \in (0, T) \times \mathcal{D}$. Hence, integrating on $[0, T]$, we get

$$\begin{aligned} & \left(\int_0^T (\frac{2}{T})^{12} \exp(-se^{2\lambda|\psi|_\infty} \theta) dt \right) \left(\mathbb{E} \int_{\mathcal{D}} p_i^2(T, x) dx \right) \\ & \leq e^{CT} \mathbb{E} \left(\int_0^T \int_{\mathcal{D}} \theta^3(t) e^{s\alpha(t,x)} p_i^2(s, x) dx dt \right) \end{aligned}$$

Thus

$$\mathbb{E} \int_{\mathcal{D}} p_i^2(T, x) dx \leq e^{CT} \mathbb{E} \left(\int_0^T \int_{\mathcal{D}} \theta^3(t) e^{s\alpha(t,x)} p_i^2(s, x) dx dt \right)$$

Therefore, taking into account the estimate (34), we get

$$\begin{aligned} \mathbb{E} \left(\sum_{i=1}^m \int_{\mathcal{D}} p_i^2(T, x) dx \right) & \leq e^{CT} \mathbb{E} \left(\sum_{i=1}^m \int_0^T \int_{\mathcal{D}} \theta^3(t) e^{s\alpha(t,x)} p_i^2(s, x) dx dt \right) \\ & \leq C' \mathbb{E} \left(s^\ell \int_{G_0 \times (0, T)} e^{2s\alpha\theta^\ell} |p_1|^2 dx dt \right), \end{aligned}$$

where $C' = e^{CT} \tilde{C}_0$. $\square$

Therefore, the unique continuation of the forward stochastic degenerate equation (4.1) is an immediate consequence of the Proposition 4.3.

Corollary 4.4. *Let p be a solution of (4.1) satisfying $p(t, x) = 0$ $\mathbb{P}$ -a.s., for all $t \in [0, T]$ and almost every $x \in G$. Then, $p(t, x) = 0$ $\mathbb{P}$ -a.s., for all $t \in [0, T]$ and almost every $x \in \mathcal{D}$.*

Notice that to establish the proof of the Theorem 1.2, we will need the following well-known functional analysis lemma (See [18, Theorem 2.2, p. 208]).

Lemma 4.5. *Let X, Y, Z be three Hilbert spaces, X^*, Y^*, Z^* their dual spaces and $F \in \mathcal{L}(X, Z)$, $G \in \mathcal{L}(Y, Z)$. Assume that Y is separable. Then $\text{Range}(F) \subset \text{Range}(G)$ if and only if there exists a constant $C > 0$ such that*

$$\|F^*z\|_{X^*} \leq C\|G^*z\|_{Y^*}, \quad z \in Z^*,$$

where F^* and G^* are the adjoint operators.

Arguing like in [6, 7] for the case of one equation, we have the following proof

Proof of the Theorem 1.2. Let us consider the following operators

$$\begin{aligned} S_0 : \quad L^2(\Omega, \mathcal{F}_T, L^2(\mathcal{D})^m) &\longrightarrow L^2(\mathcal{D})^m \\ \eta &\longmapsto y^{0,\eta}(0, x), \end{aligned} \quad (39)$$

where $y^{0,\eta}$ is the solution of (1.1) with final datum η and control $v = 0$, and

$$\begin{aligned} L_0 : \quad L^2_{\mathcal{F}}(\Omega, L^2(0, T; L^2(\mathcal{D}))) &\longrightarrow L^2(\mathcal{D})^m \\ v &\longmapsto y^{v,0}(0, x), \end{aligned} \quad (40)$$

where $y^{v,0}$ is the solution of (1.1) with control v and final datum $\eta = 0$. Hence, Theorem 1.2 holds if $\text{Range}(S_0) \subset \text{Range}(L_0)$. Let denote by H the space $H = L^2(\mathcal{D})^m$ and $\langle f, g \rangle_H = \int_H fg dx$. Let y be a solution of (1.1) and p a solution of (4.1), differentiating by Itô rule $d_s \langle y(s), p(s) \rangle_H$, integrating in $[0, T]$ and computing the mean value we obtain

$$\mathbb{E} \langle p(T), y^{v,\eta}(T) \rangle_H - \mathbb{E} \langle p(0), y^{v,\eta}(0) \rangle_H = \mathbb{E} \int_0^T \langle B \mathbb{1}_G v, p(s) \rangle_H ds. \quad (41)$$

Consequently, for all $p_0 \in H$,

$$(S_0^* p_0)(x) = p(T, x) \quad \text{and} \quad (L_0^* p_0)(t, x) = -\mathbb{1}_G B^* p(t, x).$$

Proposition 4.3 implies that there exists a constant $C > 0$ such that

$$|S_0^* p_0|_{L^2(\Omega, \mathcal{F}_T, L^2(\mathcal{D})^m)} \leq C |L_0^* p_0|_{L^2_{\mathcal{F}}(\Omega, L^2(0, T; L^2(\mathcal{D})))}.$$

By Lemma 4.5, we get $\text{Range}(S_0) \subset \text{Range}(L_0)$ which completes the proof. $\square$

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0.3.3 Controllability of two stochastic backward parabolic heat equations coupled by their second state variable under one control force

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Controllability of two stochastic backward parabolic heat equations coupled by their second state variable under one control force

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Abstract:	In this paper we study the null controllability of two backward stochastic heat equations coupled by their second variable state and governed by one control force.

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ABSTRACT. In this paper we study the null controllability of two backward stochastic heat equations coupled by their second variable state and governed by one control force.

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$L_{\mathcal{F}}^{\infty}(0, T; H)$ the Banach space consisting of all H -valued $\{\mathcal{F}_t\}_{t \geq 0}$ -adapted bounded processes; by $L_{\mathcal{F}}^2(\Omega; C([0, T]; H))$ the Banach space consisting of all H -valued $\{\mathcal{F}_t\}_{t \geq 0}$ -adapted processes $X(\cdot)$ such that $\mathbb{E} \left(\|X(\cdot)\|_{C(0, T; H)}^2 \right) < \infty$, with the canonical norm. The null controllability of system (1.1) is formulated as follows:

Definition 1.1. *System (1.1) is said to be null controllable at time $T > 0$ if for any given $y_T = (y_{1T}, y_{2T}) \in L^2(\Omega, \mathcal{F}_T, \mathbb{P}; (L^2(D))^2)$, one can find a control $v \in L_{\mathcal{F}}^2(0, T; L^2(G))$ such that the solution $((y_1, Y_1), (y_2, Y_2))$ of system (1.1) satisfies $y_1(0) = y_2(0) = 0$ in D , $\mathbb{P}$ -a.s.*

In the last decades, many results on controllability of equations and coupled systems have been established. In deterministic case M. Gonzalez-Burgos, L. de Teresa [5] provided a null controllability result for a cascade parabolic system, F. Ammar-Khodja et al. [1] obtained several results characterizing the null controllability of fully coupled systems with m -control forces by a generalized Kalman rank condition.

In stochastic case, the controllability results for one equation were initiated by Barbu, Răscanu and Tessitore [2], later others results were obtained by Tang and Zhang [8] and Hernández-Santamaría et al. [3] and references therein. Whereas, there are only few results relating to coupled systems, among which, Hongheng Li and Qi Lu [6] studied a system of two backward stochastic equations coupled by their first variable state under one control force. In [4] the author obtained controllability results for cascade system of backward stochastic equations knowing that all coefficients of the sub-diagonal in the coupling matrix A are non-zero.

In both cases, deterministic and stochastic, the key ingredient for proving the null controllability is to obtain the observability inequality for the corresponding adjoint system. The observability inequality is basically obtained by means of estimates which are called Carleman estimates.

In the current paper, we are interested in a system of two backward coupled parabolic equations coupled by their second variable state under the action of one control force.

Now, under the condition (1.2) on the second coupling matrix C , our main result about the controllability of the system (1.1) is the following

Theorem 1.2. *Assume that the second coupling matrix C satisfies the assumption (1.2), given $y_T = (y_{1T}, y_{2T}) \in L^2(\Omega, \mathcal{F}_T, \mathbb{P}; (L^2(D))^2)$. There exist a control $v \in L_{\mathcal{F}}^2(0, T; L^2(G))$ such that the solution $((y_1, Y_1), (y_2, Y_2))$ of system (1.1) satisfies $y_1(0) = y_2(0) = 0$ in D , $\mathbb{P}$ -a.s.*

Before any computations, let's discuss the well-posedness of the systems concerned by this paper, indeed, in the light of the results established for a single equation (see [2]) we therefore infer the following results on systems of coupled forward/backward stochastic equations

$$\begin{cases} d_t y(t, x) = I_2 \Delta y(t, x) + A(t, x) y(t, x) dt + F(t, x) dt + b(t, x) y(t, x) dW(t) & \text{in } Q, \\ y(t, x) = 0 & \text{on } \Sigma, \\ y(0, x) = y_0(x) & \text{in } \mathcal{D}, \end{cases} \quad (1.3)$$

where

$$\begin{aligned} (i) \quad & F \in L_{\mathcal{F}}^2(\Omega; L^2([0, T]; L^2(\mathcal{D})))^2, \\ (ii) \quad & A \in L_{\mathcal{F}}^{\infty}(\Omega; L^{\infty}([0, T]; L^{\infty}(\mathcal{D})))^{2 \times 2}, \quad b \in L_{\mathcal{F}}^{\infty}(\Omega; L^{\infty}([0, T]; L^{\infty}(\mathcal{D}))) \\ (iii) \quad & y_0 \in (L^2(\mathcal{D}))^2. \end{aligned} \quad (1.4)$$

Under the assumptions (1.4) equation (1.3) has a unique solution y belonging to $(L_{\mathcal{F}}^2(\Omega; C([0, T]; L^2(\mathcal{D}))))^2$. Moreover, if $y_0 \in (H_0^1(\mathcal{D}))^2$ and $b \in L_{\mathcal{F}}^{\infty}(\Omega; L^{\infty}([0, T]; W^{1, \infty}(\mathcal{D})))$ then $y(t)$ is in $(H_0^1(\mathcal{D}) \cap H^2(\mathcal{D}))^2$, $\mathbb{P}$ -almost surely for every $t \in]0, T]$.

As for the backward stochastic equation

$$\begin{cases} dp(t, x) = [-I_2 \Delta p(t, x) - Ap(t, x) + b(t, x)k(t, x)dt + G(t, x)] dt + k(t, x)dW(t) & \text{in } Q, \\ p(t, x) = 0 & \text{on } \Sigma, \\ p(T, x) = p_T(x) & \text{in } \mathcal{D}, \end{cases} \quad (1.5)$$

where

$$\begin{aligned} (i) \quad & G \in \left(L^2_{\mathcal{F}}(\Omega; L^2(0, T; L^2(\mathcal{D}))) \right)^2, \\ (ii) \quad & p_T \in \left(L^2(\Omega, \mathcal{F}_T, \mathbb{P}; L^2(\mathcal{D})) \right)^2. \end{aligned} \quad (1.6)$$

Under the previous conditions, the backward stochastic differential system (1.5) has a unique solution (p, k) with $p \in \left(L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D}))) \cap L^2_{\mathcal{F}}(\Omega; L^2([0, T]; H^1_0(\mathcal{D}))) \right)^2$ and $k \in \left(L^2_{\mathcal{F}}(\Omega; L^2([0, T]; L^2(\mathcal{D}))) \right)^2$.

Moreover, if $p_T \in \left(L^2(\Omega, \mathcal{F}_T, \mathbb{P}; H^1_0(\mathcal{D})) \right)^2$ and $b \in L^{\infty}_{\mathcal{F}}(\Omega, L^{\infty}([0, T], W^{1,\infty}(\mathcal{D})))$, then we have : $p \in \left(L^2(\Omega; L^2([0, T]; H^2(\mathcal{D}))) \cap L^2(\Omega; C([0, T]; H^1_0(\mathcal{D}))) \right)^2$ and $k \in \left(L^2(\Omega; L^2([0, T]; H^1_0(\mathcal{D}))) \right)^2$.

The rest of this paper is organized as follows : section 2 is devoted Carleman estimates and the observability inequality for the adjoint system (2.4) associated to the system (1.1). And as an immediate consequence of the previous computations we infer the unique continuation property for the adjoint system (2.4). In section 3, we give the proof of the main theorem 1.2. And finally in the appendix we establish a Caccioppoli type inequality for a forward stochastic parabolic equation.

All along the article, we use generic constants for the estimates, whose values may change from line to line.

2. OBSERVABILITY INEQUALITY FOR THE ADJOINT SYSTEM ASSOCIATED TO (1.1)

First of all, let us introduce the weighted functions necessary for the Carleman estimation, for this we recall the following result (see [8, Lemma 5.1])

Lemma 2.1. *Let $G_0 \Subset G$ be an arbitrary fixed subdomain of $\mathcal{D}$. Then there exists a function $\psi \in C^4(\overline{\mathcal{D}})$ such that*

$$\psi(x) > 0 \quad \forall x \in \mathcal{D}, \quad \psi|_{\partial\Omega} = 0, \quad |\nabla\psi(x)| > 0 \quad \forall x \in \overline{\mathcal{D}} \setminus G_0.$$

Put

$$\theta(t) = \frac{1}{t^2(T-t)^2}, \quad \beta(x) = e^{\lambda\psi(x)} - e^{2\lambda|\psi|_{\infty}}, \quad \alpha(t, x) = \theta(t)\beta(x), \quad \text{and} \quad \varphi(t, x) = \theta(t)e^{\lambda\psi(x)} \quad (2.1)$$

Therefore, we have the following estimate of forward stochastic heat equations (see [7, Theorem 1.1]).

Lemma 2.2. *For any $T > 0$, there exists a constant $\lambda_0 > 0$ such that for all $\lambda \geq \lambda_0$, one can find two constants $C = C(\lambda) > 0$ and $s_0 = s_0(\lambda) > 0$ so that for all $p \in L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(\mathcal{D}))) \cap L^2_{\mathcal{F}}(0, T; H^1_0(\mathcal{D}))$, $f_i \in L^2_{\mathcal{F}}(0, T; L^2(\mathcal{D}))$ ($i = 1, 2$) satisfying*

$$dp - \Delta p \, dt = f_1 \, dt + f_2 \, dW(t) \quad (2.2)$$

it holds that

$$\begin{aligned} \mathbb{E} \int_Q e^{2s\alpha} (s^3 \theta^3 p^2 + s \theta |\nabla p|^2) \, dx \, dt \leq C \left(\mathbb{E} \int_0^T \int_{G_1} s^3 \theta^3 e^{2s\alpha} p^2 \, dx \, dt \right. \\ \left. + \mathbb{E} \int_Q e^{2s\alpha} f_1^2 \, dx \, dt + \mathbb{E} \int_Q s^2 \theta^2 f_2^2 e^{2s\alpha} \, dx \, dt \right) \end{aligned} \quad (2.3)$$

for any $s \geq s_0$.

The adjoint system of (1.1) is the following

$$\begin{cases} dp_1 = (\Delta p_1 - a_1 p_1) \, dt - (c_{1,1} p_1 + c_{2,1} p_2) \, dW(t), & \text{in } Q \\ dp_2 = (\Delta p_2 - a_2 p_2) \, dt - (c_{1,2} p_1 + c_{2,2} p_2) \, dW(t), & \text{in } Q \\ p_i = 0 \quad i \in \{1, 2\} & \text{on } \Sigma, \\ p_i(0, \cdot) = p_{i0}(\cdot) \quad i \in \{1, 2\} & \text{in } \mathcal{D}, \end{cases} \quad (2.4)$$

Observe that we deal with particular equations involved in (2.4), thus using Lemma 2.2 we infer the following result

Corollary 2.3. *There exists a constant $\lambda_0 > 0$ such that for all $\lambda \geq \lambda_0$, one can find two constants $C = C(\lambda) > 0$ and $s_0 = s_0(\lambda) > 0$ so that for all $p_i \in L^2_{\mathcal{F}}(\Omega; C([0, T]; L^2(G))) \cap L^2_{\mathcal{F}}(0, T; H^1_0(G))$, and $f \in L^2_{\mathcal{F}}(0, T; L^2(G))$ satisfying*

$$dp_i = (\Delta p_i - a p_i) \, dt - (\alpha p_1 + \beta p_2) \, dW(t)$$

it holds that

$$\mathbb{E} \int_Q e^{2s\alpha} (s^3 \theta^3 p_i^2 + s\theta |\nabla p_i|^2) dxdt \leq C \left(\mathbb{E} \int_0^T \int_{G_1} s^3 \theta^3 e^{2s\alpha} p_i^2 dxdt + \sum_{j=1}^2 \mathbb{E} \int_Q s^2 \theta^2 p_j^2 e^{2s\alpha} dxdt \right) \quad (2.5)$$

for any $s \geq s_0$.

We have the following intermediate estimate for the problem (2.4).

Proposition 2.4. *Assume that the second coupling matrix C satisfies the assumption (1.2), there exist two positive constants C and s_0 such that, for every $p_0 = (p_{10}, p_{20})^* \in L^2(Q)^2$, the solution $p = (p_1, p_2)^*$ to (2.4) satisfies*

$$\sum_{i=1}^2 \mathbb{E} \int_Q ((s\theta)^3 p_i^2 + s\theta |\nabla p_i|^2) e^{2s\alpha} dxdt \leq C \mathbb{E} \int_0^T \int_G (s\theta)^k e^{2s\alpha} p_1^2 dxdt. \quad (2.6)$$

for all $s \geq s_0$.

Proof. Applying respectively the estimate (2.5) to both the equations of the system (2.4) we get

$$\begin{aligned} & \sum_{i=1}^2 \mathbb{E} \int_Q ((s\theta)^3 p_i^2 + s\theta |\nabla p_i|^2) e^{2s\alpha} dxdt \\ & \leq C \left\{ \sum_{i=1}^2 \mathbb{E} \int_0^T \int_G (s\theta)^3 e^{2s\alpha} p_i^2 dxdt + \sum_{j=1}^2 \mathbb{E} \int_Q (s\theta)^2 p_j^2 e^{2s\alpha} dxdt \right\}. \end{aligned} \quad (2.7)$$

For s enough large the last term in the right hand-side of (2.7) can be absorbed by terms in the left hand-side, hence, we get

$$\sum_{i=1}^2 \mathbb{E} \int_Q ((s\theta)^3 p_i^2 + s\theta |\nabla p_i|^2) e^{2s\alpha} dxdt \leq C \left\{ \sum_{i=1}^2 \mathbb{E} \int_0^T \int_G (s\theta)^3 e^{2s\alpha} p_i^2 dxdt \right\}. \quad (2.8)$$

The next task is to absorb the term $\mathbb{E} \int_0^T \int_G (s\theta)^3 e^{2s\alpha} p_2^2 dxdt$ from the estimate (2.8). Indeed, let consider $G_1 \subset \mathcal{O}_1 \Subset \mathcal{O}_0 \subset G_0$ and $\xi \in C^\infty(\mathbb{R}^N)$ such that $0 \leq \xi \leq 1$ in $\mathbb{R}^N$ $\xi = 1$ in $\mathcal{O}_1$ and

$$\text{supp}(\xi) \subset \mathcal{O}_0, \quad \frac{\Delta \xi}{\xi^{1/2}} \in L^\infty(\mathcal{D}), \quad \frac{\nabla \xi}{\xi^{1/2}} \in L^\infty(\mathcal{D})^N. \quad (2.9)$$

We also denote by $u = (s\theta)^3 e^{2s\alpha}$. Differentiating $d \left(\int_{\mathcal{O}_0} \xi u p_1 p_2 dx \right)$ integrating on $[0, T]$ and computing the mean value we get

$$\begin{aligned} c_{21} c_{22} \mathbb{E} \int_0^T \int_{\mathcal{O}_1} (s\theta)^3 e^{2s\alpha} p_2^2 dxdt & \leq c_{21} c_{22} \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^3 \xi e^{2s\alpha} p_2^2 dxdt \\ & \leq \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (\xi d u p_1 p_2) dxdt + \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi u (p_1 \Delta p_2 + p_2 \Delta p_1) dxdt \\ & \quad + \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi u (c_{11} c_{22} + c_{21} c_{12} - a_1 - a_2) p_1 p_2 dxdt + \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi u c_{11} c_{12} p_1^2 dxdt \\ & \leq I_1 + I_2 + I_3 + I_4. \end{aligned} \quad (2.10)$$

Since $du = 3s^3 \theta^2 \dot{\theta} e^{2s\alpha} dt + 2s^4 \theta^3 \dot{\theta} \beta e^{2s\alpha} dt$, thus, we infer $|\frac{du}{dt}| \leq C s^4 \theta^{9/2} e^{2s\alpha}$ therefore, for $\varepsilon > 0$, we have

$$\begin{aligned} I_1 & \leq C \mathbb{E} \int_0^T \int_{\mathcal{O}_0} s^4 \theta^{9/2} \xi e^{2s\alpha} p_1 p_2 dxdt \\ & \leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^3 \xi e^{2s\alpha} p_2^2 dxdt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} s^5 \theta^6 \xi e^{2s\alpha} p_1^2 dxdt \end{aligned} \quad (2.11)$$

Thanks to Green formula, we get

$$\begin{aligned} I_2 & = \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi u (p_1 \Delta p_2 + p_2 \Delta p_1) dxdt \\ & = -\mathbb{E} \int_0^T \int_{\mathcal{O}_0} p_1 \nabla(\xi u) \nabla p_2 dxdt - \mathbb{E} \int_0^T \int_{\mathcal{O}_0} p_2 \nabla(\xi u) \nabla p_1 dxdt - 2\mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi u \nabla p_1 \nabla p_2 dxdt \\ & = J_1 + J_2 + J_3. \end{aligned}$$

We have $\nabla(\xi u) = (s\theta)^3 \nabla \xi e^{2s\alpha} + 2(s\theta)^4 \xi \nabla \beta e^{2s\alpha}$, thus

$$J_1 = -\mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^3 e^{2s\alpha} p_1 \nabla \xi \nabla p_2 dx dt - \mathbb{E} \int_0^T \int_{\mathcal{O}_0} 2(s\theta)^4 \xi e^{2s\alpha} p_1 \nabla \beta \nabla p_2 dx dt.$$

Using Young-inequality and assumptions on ξ (2.9) we get

$$|J_1| \leq C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^7 e^{2s\alpha} p_1^2 dx dt + \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} s\theta \xi e^{2s\alpha} |\nabla p_2|^2 dx dt$$

With similar the computations made for J_1 we get

$$J_2 = -\mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^3 e^{2s\alpha} p_2 \nabla \xi \nabla p_1 dx dt - \mathbb{E} \int_0^T \int_{\mathcal{O}_0} 2(s\theta)^4 \xi e^{2s\alpha} p_2 \nabla \beta \nabla p_1 dx dt$$

Therefore,

$$|J_2| \leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^3 e^{2s\alpha} p_2^2 dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^5 e^{2s\alpha} |\nabla p_1|^2 dx dt,$$

and

$$|J_3| \leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi s\theta |\nabla p_2|^2 e^{2s\alpha} dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^5 e^{2s\alpha} |\nabla p_1|^2 dx dt.$$

Consequently, we infer

$$\begin{aligned} |I_2| &\leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi (s\theta)^3 e^{2s\alpha} p_2^2 dx dt + \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} s\theta \xi e^{2s\alpha} |\nabla p_2|^2 dx dt \\ &\quad + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^7 e^{2s\alpha} p_1^2 dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^5 e^{2s\alpha} |\nabla p_1|^2 dx dt. \end{aligned} \quad (2.12)$$

The first term in the right hand-side of (2.12) can be absorbed by the term in the right hand-side of (2.10). The remaining terms are bounded as follows

$$|I_3| \leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi (s\theta)^3 p_2^2 e^{2s\alpha} dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^3 e^{2s\alpha} p_1^2 dx dt. \quad (2.13)$$

and

$$|I_4| \leq C \mathbb{E} \int_0^T \int_{\mathcal{O}_0} \xi (s\theta)^3 p_1^2 e^{2s\alpha} dx dt. \quad (2.14)$$

Now, from (2.11)-(2.14) altogether we infer

$$\begin{aligned} \mathbb{E} \int_0^T \int_{\mathcal{O}_1} (s\theta)^3 e^{2s\alpha} p_2^2 dx dt &\leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} s\theta \xi e^{2s\alpha} |\nabla p_2|^2 dx dt \\ &\quad + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^7 e^{2s\alpha} p_1^2 dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{O}_0} (s\theta)^5 e^{2s\alpha} |\nabla p_1|^2 dx dt. \end{aligned} \quad (2.15)$$

Replacing the previous result in (2.8), the first term in the right hand-side of (2.15) can be absorbed by the term in the left hand-side of (2.8), whereas the last term in (2.15) can be bounded by $C \mathbb{E} \int_0^T \int_{G_0} (s\theta)^7 e^{2s\alpha} p_1^2 dx dt$ using the Caccioppoli inequality, this completes the proof of (2.6). $\square$

Next, we establish an observability inequality for the system (2.4). Such inequality is of a great interest in the proof of the main theorem as well as in the proof of the unique continuation of the adjoint problem which is a cascade system of coupled forward stochastic equations (2.4).

Theorem 2.5. *Under assumptions on the second coupling matrix C , there exist two positive constants C and s_0 such that, for every $p_0 = (p_{10}, p_{20})^* \in L^2(Q)^2$, the solution $p = (p_1, p_2)^*$ to (2.4) satisfies*

$$\mathbb{E} \left(\int_{\mathcal{D}} p_1^2(T, x) dx + \int_{\mathcal{D}} p_2^2(T, x) dx \right) \leq C \mathbb{E} \left(s^\ell \int_{G_0 \times (0, T)} e^{2s\alpha} \theta^\ell |p_1|^2 dx dt \right), \quad (2.16)$$

for all $s \geq s_0$.

Proof. For $i = 1, 2$, differentiating $d_\tau \|p_i\|_{L^2(0,1)}^2$, integrating on $[s, t]$ (with $0 \leq s \leq t \leq T$) and taking the mean value, we have

$$\mathbb{E} \int_{\mathcal{D}} p_i^2(t, x) dx \leq \mathbb{E} \int_{\mathcal{D}} p_i^2(s, x) dx + C \mathbb{E} \int_s^t \int_{\mathcal{D}} p_i^2(\ell, x) dx d\ell.$$

Tanks to Gronwall's lemma, we get

$$\mathbb{E} \int_{\mathcal{D}} p_i^2(t, x) dx \leq e^{C(t-s)} \mathbb{E} \int_{\mathcal{D}} p_i^2(s, x) dx \leq e^{CT} \mathbb{E} \int_{\mathcal{D}} p_i^2(s, x) dx.$$

Knowing that $\theta^3(t) e^{2s\alpha(t,x)} \geq (\frac{2}{T})^{12} \exp(-se^{2\lambda|\psi|_\infty} \theta)$ for all $(t, x) \in (0, T) \times \mathcal{D}$. Hence, integrating on $[0, T]$, we get

$$\left(\int_0^T (\frac{2}{T})^{12} \exp(-se^{2\lambda|\psi|_\infty} \theta) dt \right) \left(\mathbb{E} \int_{\mathcal{D}} p_i^2(T, x) dx \right) \leq e^{CT} \mathbb{E} \left(\int_0^T \int_{\mathcal{D}} \theta^3(t) e^{2s\alpha(t,x)} p_i^2(s, x) dx dt \right)$$

Thus

$$\mathbb{E} \int_{\mathcal{D}} p_i^2(T, x) dx \leq e^{CT} \mathbb{E} \left(\int_0^T \int_{\mathcal{D}} \theta^3(t) e^{2s\alpha(t,x)} p_i^2(s, x) dx dt \right)$$

Therefore, taking into account the estimate (2.6), we get

$$\begin{aligned} \mathbb{E} \left(\int_{\mathcal{D}} p_1^2(T, x) dx + \int_{\mathcal{D}} p_2^2(T, x) dx \right) &\leq e^{CT} \mathbb{E} \left(\sum_{i=1}^2 \int_0^T \int_{\mathcal{D}} \theta^3(t) e^{2s\alpha(t,x)} p_i^2(s, x) dx dt \right) \\ &\leq C' \mathbb{E} \left(s^\ell \int_{G_0 \times (0, T)} e^{2s\alpha} \theta^\ell |p_1|^2 dx dt \right), \end{aligned}$$

for $s > s_0$ and where $C' = e^{CT} \widetilde{C}_0$. □

Straightway, we get the following weak controllability result for the adjoint system (2.4). As we know, both the equations of the system (2.4) are only coupled in their noise part, and yet the unique continuation property is fulfilled for such system.

Corollary 2.6 (Unique Continuation of the adjoint system (2.4)). *Let (p_1, p_2) be a solution of (2.4) satisfying $p_1(t, x) = 0$ $\mathbb{P}$ -a.s., for all $t \in [0, T]$ and almost every $x \in G$. Then, $p_1(t, x) = p_2(t, x) = 0$ $\mathbb{P}$ -a.s., for all $t \in [0, T]$ and almost every $x \in \mathcal{D}$.*

3. PROOF OF THE MAIN RESULT

Proof of the Theorem 1.2. Let us consider the following operators

$$\begin{aligned} S_0 : \quad L^2(\Omega, \mathcal{F}_T, L^2(\mathcal{D})^2) &\longrightarrow L^2(\mathcal{D})^2 \\ \eta &\longmapsto y^{0, \eta}(0, x), \end{aligned} \tag{3.1}$$

where $y^{0, \eta}$ is the solution of (1.1) with final datum η and control $v = 0$, and

$$\begin{aligned} L_0 : \quad L^2_{\mathcal{F}}(\Omega, L^2(0, T; L^2(\mathcal{D}))) &\longrightarrow L^2(\mathcal{D})^2 \\ v &\longmapsto y^{v, 0}(0, x), \end{aligned} \tag{3.2}$$

where $y^{v, 0}$ is the solution of (1.1) with control v and final datum $\eta = 0$. Hence, Theorem 1.2 holds if $\text{Range}(S_0) \subset \text{Range}(L_0)$. Let denote by H the space $H = L^2(\mathcal{D})^2$ and $\langle f, g \rangle_H = \int_H f g dx$. Let $y = (y_1, y_2)$ be a solution of (1.1) and $p = (p_1, p_2)$ a solution of (2.4), differentiating by Itô rule $d_s \langle y(s), p(s) \rangle_H$, integrating in $[0, T]$ and computing the expectation we obtain

$$\mathbb{E} \langle p(T), y^{v, \eta}(T) \rangle_H - \mathbb{E} \langle p(0), y^{v, \eta}(0) \rangle_H = \mathbb{E} \int_0^T \langle B \mathbb{1}_G v, p(s) \rangle_H ds. \tag{3.3}$$

Consequently, for all $p_0 \in H$,

$$(S_0^* p_0)(x) = p(T, x) \quad \text{and} \quad (L_0^* p_0)(t, x) = -\mathbb{1}_G B^* p(t, x).$$

Theorem 2.5 implies that there exists a constant $C > 0$ such that

$$|S_0^* p_0|_{L^2(\Omega, \mathcal{F}_T, L^2(\mathcal{D})^2)} \leq C |L_0^* p_0|_{L^2_{\mathcal{F}}(\Omega, L^2(0, T; L^2(\mathcal{D})))}.$$

By [9, Theorem 2.2, p. 208], we get $\text{Range}(S_0) \subset \text{Range}(L_0)$ which completes the proof. □

4. APPENDIX

In this section, we establish a Caccioppoli type inequality for a forward stochastic parabolic equation, indeed, consider

$$dp = \Delta p dt + ap dt + f dt + g dW(t) \quad (4.1)$$

And $\mathcal{U}_0, \mathcal{U}_1$ two nonempty open subsets verifying

$$\mathcal{U}_1 \Subset \mathcal{U}_0 \Subset G \subset \mathcal{D},$$

and η be a smooth function $\eta \in \mathcal{C}^\infty(\mathbb{R}^n, \mathbb{R})$ such that $0 \leq \eta \leq 1$ in $\mathbb{R}^n$ $\eta = 1$ in $\mathcal{U}_1$ and

$$\text{supp} \eta \subset \mathcal{U}_0, \quad \frac{\nabla \eta}{\eta^{1/2}} \in L^\infty(\mathcal{D})^n. \quad (4.2)$$

Therefore, we have the following result

Lemma 4.1. *There exists a positive constant C such that the solution p of the equation (4.1) satisfies*

$$\mathbb{E} \int_0^T \int_{\mathcal{U}_1} |\nabla p|^2 e^{2s\alpha} dx dt \leq C \left(\mathbb{E} \int_0^T \int_{\mathcal{U}_0} s^2 \theta^2 p^2 e^{2s\alpha} dx dt + \mathbb{E} \int_0^T \int_{\mathcal{U}_0} f^2 e^{2s\alpha} dx dt + \mathbb{E} \int_0^T \int_{\mathcal{U}_0} g^2 e^{2s\alpha} dx dt \right) \quad (4.3)$$

Proof. Let p be a solution of (4.1), differentiating $d \int_{\mathcal{U}_0} \eta p^2 e^{2s\alpha} dx$, integrating on $[0, T]$ and computing the mean value we get

$$\begin{aligned} 0 = & -2\mathbb{E} \int_0^T \int_{\mathcal{U}_0} \nabla \eta \nabla p p e^{2s\alpha} dx dt - 2\mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta |\nabla p|^2 e^{2s\alpha} dx dt - 4\mathbb{E} \int_0^T \int_{\mathcal{U}_0} s \eta p \nabla p \nabla \alpha e^{2s\alpha} dx dt \\ & + 2\mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta a p^2 e^{2s\alpha} dx dt + 2\mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta f p e^{2s\alpha} dx dt + 2\mathbb{E} \int_0^T \int_{\mathcal{U}_0} s \eta \alpha_t p^2 e^{2s\alpha} dx dt + 2\mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta g^2 e^{2s\alpha} dx dt. \end{aligned} \quad (4.4)$$

Thus,

$$\begin{aligned} \mathbb{E} \int_0^T \int_{\mathcal{U}_1} |\nabla p|^2 e^{2s\alpha} dx dt & \leq \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta |\nabla p|^2 e^{2s\alpha} dx dt \\ & \leq \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \nabla \eta \nabla p p e^{2s\alpha} dx dt + \mathbb{E} \int_0^T \int_{\mathcal{U}_0} s \eta p \nabla p \nabla \alpha e^{2s\alpha} dx dt \\ & \quad + \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta f p e^{2s\alpha} dx dt + \mathbb{E} \int_0^T \int_{\mathcal{U}_0} (\eta a + s \eta \alpha_t) p^2 e^{2s\alpha} dx dt + \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta g^2 e^{2s\alpha} dx dt \\ & \leq J_1 + J_2 + J_3 + J_4 + J_5. \end{aligned} \quad (4.5)$$

For $\varepsilon > 0$ and using Young inequality we get

$$J_1 \leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta |\nabla p|^2 e^{2s\alpha} dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{U}_0} p^2 e^{2s\alpha} dx dt \quad (4.6)$$

$$J_2 \leq \varepsilon \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta |\nabla p|^2 e^{2s\alpha} dx dt + C_\varepsilon \mathbb{E} \int_0^T \int_{\mathcal{U}_0} s^2 \theta^2 p^2 e^{2s\alpha} dx dt \quad (4.7)$$

$$J_3 \leq \frac{1}{2} \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta f^2 e^{2s\alpha} dx dt + \frac{1}{2} \mathbb{E} \int_0^T \int_{\mathcal{U}_0} \eta p^2 e^{2s\alpha} dx dt \quad (4.8)$$

And the last term

$$J_4 \leq C \mathbb{E} \int_0^T \int_{\mathcal{U}_0} s^2 \theta^2 p^2 e^{2s\alpha} dx dt \quad (4.9)$$

From (4.5)–(4.9) all together we infer the desired estimate (4.3). $\square$

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0.4 Activités Pédagogiques

0.4.1 Cursus Professionnel :

1995–2000 : *Enseignant du 2ieme cycle de mathématiques au lycée Ouaouizerth .*
Ouaouizerth.

2000–2002 : *Préparation du concours d'agrégation mathématiques à l' ENS .*
Marrakech–Casablanca.

2002–2005 : *Professeur agrégé de mathématiques au collège royal préparatoire aux techniques aéronautiques BEFRA.* Marrakech.

2005–2009 : *Professeur agrégé de mathématiques au Lycée Zerktouni.* Marrakech.
Pendant 3 ans j'ai été **maître d'application**¹ pour les étudiants enseignants de
l'ENS

2012–2019 : *Professeur agrégé de mathématiques au collège royal préparatoire aux techniques aéronautiques BEFRA.* Marrakech.

2009–2012 : *Professeur agrégé de mathématiques au **BTS** Lycée Mohamed 6 .*
Marrakech.

2019– Maintenant : *Professeur assistant au département de mathématiques de l'Ecole normale supérieure, Université Cadi Ayyad.* Marrakech.

0.4.2 Encadrement

1) Mémoire de fin d'étude des étudiants Imrharn Wiame et Faouzi Soumia intitulé : *Équations différentielles $x' = f(t, x)$ exemples et applications.* 2021. École Normale Supérieure Marrakech.

2) Mémoire de fin d'étude des étudiants Badiha Hafsa et Auhmani Yasmine intitulé : *Initiation au logiciel Python.* 2022. École Normale Supérieure Marrakech.

3) Mémoire de fin d'étude des étudiants Bouswir Malika et Boudad Hasna intitulé *Recherche de zéro d'une fonction réelle.* 2022. École Normale Supérieure Marrakech.

0.4.3 Modules enseignés a l'ENS

Mathématiques pour licence d'enseignement Sciences de la vie et la terre

Chapitre I : Rappels.

1 - Point et vecteur.

¹Les étudiants professeurs de l'ENS passent leur stage dans mes classes et sous mon encadrement.

2 - Droite et plan.

Chapitre II : Calcul matriciel.

1 - Définition d'une matrice.

2 - Opérations algébriques sur les Matrices. 3 - Matrice inversible.

Chapitre III : Déterminant d'une matrice carrée.

1 - Définition et calcul d'un déterminant.

2 - Caractérisation d'une matrice inversible.

Chapitre IV : Résolution des systèmes linéaires.

1 - Système de m équations à n inconnues.

2 - Nature d'un système et systèmes équivalents.

3 - Résolution par la méthode de Gauss.

4 - Résolution d'un système de Cramer.

Chapitre V : Suites réelles.

1 - Définition générale.

2 - Convergence d'une suite.

3 - Suites Particulières.

Chapitre VI : Limites et continuité d'une fonction.

1 - Limite d'une fonction quand x tend vers une Valeur (finie ou infinie).

2 - Opérations sur les limites.

3 - Continuité en un point.

4 - Prolongement Par continuité.

5 - Fonctions continues sur un intervalle $[a, b]$.

6 - Fonctions trigonométriques Inverses.

Chapitre VII : Dérivée d'une fonction.

1 - Dérivée en un point.

2 - Opérations sur les fonctions Dérivables.

3 - Dérivée d'une fonction réciproque.

4 - Théorème de Rolle et théorème des Accroissements finis.

5 - Règles de l'Hôpital.

Chapitre VIII : Intégrale simple.

1 - Intégrale définie.

2 - Primitive d'une fonction (intégrale Indéfinie).

3 - Méthodes d'intégration.

Chapitre IX : Equations différentielles.

1 - Généralités sur les équations différentielles.

2 - Equation différentielle du premier ordre.

3 - Equation différentielle du second ordre a coefficients Constants.

TRAVAUX DIRIGES : Exercices d'application.

Analyse 1 pour licence d'enseignement Sciences physique

1- Suites réelles

Convergence, limites, suites arithmétiques, suites géométriques, suites monotones, suites adjacentes, opérations sur les suites.

2- Fonctions numériques d'une variable réelle

Calcul des limites, continuité, théorème des valeurs intermédiaires.

3- Fonctions dérivables

Dérivée première, dérivées successives, sens de variation, Théorème de Rolle et théorème des accroissements finis

4- Fonctions convexes

Définition, fonction convexes dérivables, inégalité de convexité

5- Fonctions monotones

Définition, fonction réciproque, fonction réciproques des fonctions circulaires et des fonctions hyperboliques

6- Fonctions équivalentes et développements limités

Formule de Taylor, polynômes d'interpolation et calcul approché

7- Courbes paramétré planes

Définition, tangentes, points réguliers, points stationnaires, branches infinies, représentation en coordonnées polaires, exemples de courbes polaires.

Analyse 2 pour licence d'enseignement Sciences physique

1- Les séries

Séries numériques, séries entières, série trigonométriques et série de Fourier, critères de convergence, rayon de convergence.

2- Calcul intégral

Notion d'intégrale, calcul des primitives, intégration par partie, intégration par changement de variables, intégration des fractions rationnelles, intégral dépendant d'un paramètre.

5- Intégrale généralisée

Intégrale généralisée, critères de convergence.

3- Equations différentielles

Equations différentielles linéaire du 1er ordre, équations différentielles du 2ème ordre

4- Elément de calcul différentiel

Fonctions à plusieurs variables, dérivées partielles du 1er ordre, dérivées partielles d'ordre supérieur, fonction de classe $\mathcal{C}^1$, extremum, plan tangent à une surface dans $\mathbb{R}^3$.

5- Intégrales doubles

Intégrale double d'une fonction continue bornée, propriétés de l'intégrale double, formules de Fubini, changement de variables, extension aux intégrales triples.

6- Suites et séries de fonctions

Suites de fonctions, série de fonctions, critères de convergence, série entières, rayon de convergence, dérivation, intégration, fonctions analytiques.

Topologie pour licence d'enseignement Mathématiques

Espaces métriques

– Topologie d'un espace métrique : distance - Boules ouvertes et fermées – ouverts et fermés – voisinages – adhérence – base d'ouverts...

– Espaces métriques séparables

– Suites et applications dans les espaces métriques : continuité et continuité uniforme d'une application - valeur d'adhérence d'une suite...

– Espaces métriques compacts : Propriété de l'intersection finie – théorème de Heine...

– Espaces métriques complets : théorème du point fixe

Espaces topologiques

– Définition et exemples d'espaces topologiques : ouverts, fermés, voisinages...

– Intérieur, adhérence, frontière, point isolé, point d'accumulation

– Topologie induite et topologie quotient

– Applications continues dans les espaces topologiques

– Topologie produit

– Espaces compacts et localement compacts

– Espaces connexes

Quelques théorèmes d'analyse fonctionnelle

– Théorème de Baire et applications

– Théorème d'Ascoli

– Théorème de Stone Weierstrass

Analyse Numérique pour licence d'enseignement Mathématiques

Chapitre 1: Méthodes numériques pour la résolution de l'équation $f(x)=0$.

a) Division synthétique

b) Nombre et localisation des racines de l'équation algébrique $P(x)=0$.

- b-1) Méthode de Descartes.

- b-2) Théorème de Sturm.

c) Méthode du point fixe pour approcher les racines de $f(x)=0$.

- c-1) Théorème du point fixe.

- c-2) Majoration de l'erreur d'approximation

Chapitre 2 : Interpolation polynomiale

a) Polynôme d'interpolation de Lagrange . Existence et unicité.

b) Erreur d'interpolation

c) Différence divisée. Polynôme d'interpolation de Newton.

Chapitre 3 : Intégration numérique

a) Méthodes d'intégration de Newton-Cotes de type fermé.

b) Méthodes d'intégration de Newton-Cotes de type ouvert.

c) Erreur d'intégration.

Chapitre 4 : Systèmes linéaires

a) Décomposition $A=LU$.

- a-1) Méthode de Crout.

- a-2) Méthode de Doolittle.

- a-3) Le cas où A est symétrique.

b) Existence et unicité de la Décomposition $A=LU$.

c) Application à la résolution du système $Ax=b$ et au calcul de l'inverse de la matrice A .

Analyse Numérique pour Master Mathématiques

Ch.1 : Résolution numérique d'un système linéaire

(Méthode de Gauss ; Décomposition LU ; Méthode de Cholesky ; Méthodes de Gauss-Seidel et de Jacobi ; Relaxation).

Ch.2 : A) Résolution numérique de l'équation $f(x)=0$

(Méthode de dichotomie ; méthode des approximations successives ; méthode de Newton ; méthode de la fausse position).

Ch.2 : B) Résolution numérique de l'équation algébrique $P_n(x)=0$

(Algorithme de Horner ; Nombre de racines réelles ; Théorème de Sturm).

Ch.3 : Interpolation polynomiale

(Méthode de Lagrange ; Méthode de Newton. Etude de l'erreur).

Ch.4 : Dérivation et intégration numérique

(Méthode des trapèzes ; méthode de Simpson).

0.4.4 Supports Pédagogiques réalisé à l'ENS

- Réalisation d'un support de cours et TD pour le module : Mathématiques LE SVT S1.
- Réalisation d'un support de cours et TD pour le module : Analyse I LE PC S1.
- Réalisation d'un support de cours et TD pour le module : Analyse II LE PC S2.
- Réalisation d'un support de cours et TD pour le module : Topologie LE Maths S5.
- Réalisation d'un support de cours et TD pour le module : Analyse numérique Master de Maths S3.

0.5 Conclusion et perspectives

Ce mémoire d'habilitation universitaire est une continuation des travaux de ma thèse de doctorat sur la contrôlabilité à zéro de quelques systèmes paraboliques dégénérés, notamment le premier papier sur " La contrôlabilité d'un système d'équations paraboliques dégénérées avec une matrice de diffusion non diagonalisable". Passant ensuite aux systèmes stochastiques rétrogrades, nous avons montré que si la première matrice de couplage est cascade (triangulaire supérieure avec une sous diagonale non nul) le système peut alors être contrôlé avec un seul contrôle (voir le deuxième papier). Et finalement dans le cas de deux équations, nous avons montré qu'une condition sur la seconde matrice de couplage permet de contrôler le système avec un seul contrôle.

le contrôle des équations stochastiques est donc un champ fertile et prometteur dont nous avons l'intention d'explorer à l'avenir.

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